Andrology & Sexual Medicine

Condensed rapid revision · facts already answered in Papers 1–4 point to that answer

highlighted number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book

Chapter	Topic
Ch 66	Surgical, Radiographic and Endoscopic Anatomy of the Male Reproductive System
Ch 67	Male Reproductive Physiology
Ch 68	Integrated Men's Health: Androgen Deficiency, Cardiovascular Risk and Metabolic Syndrome
Ch 69	Male Infertility
Ch 70	Surgical Management of Male Infertility
Ch 71	Physiology of Penile Erection and Pathophysiology of ED
Ch 72	Evaluation and Management of Erectile Dysfunction
Ch 73	Priapism
Ch 74	Disorders of Male Orgasm and Ejaculation
Ch 75	Surgery for Erectile Dysfunction
Ch 76	Diagnosis and Management of Peyronie Disease
★	Putting it together: andrology decisions

66 Surgical, Radiographic and Endoscopic Anatomy of the Male Reproductive System

1 · Scrotum

Layer (superficial → deep)	Continuous with	Note
Tunica dartos	Scarpa, Colles, penile dartos	Smooth muscle
External spermatic fascia	External oblique	—
Cremasteric fascia	**Internal oblique**	**Striated** cremaster; thermoregulation
Internal spermatic fascia	Transversalis fascia	—
Tunica vaginalis (parietal, visceral)	Peritoneum	Hydrocele space; visceral layer fused to albuginea

- Anterior wall: external pudendal vessels, ilioinguinal/genitofemoral nerves; posterior: perineal vessels/nerve. Nothing crosses the raphe.
- P1 Lymphatic drainage of testis and scrotum, RPLND templates and penile cancer implications — *scrotum → ipsilateral superficial inguinal; testis → para-aortic/interaortocaval; cauda → external iliac*

KEY POINT
Scrotal lymph goes to the groin, testicular lymph to the retroperitoneum — †so scrotal violation in testis cancer adds inguinal nodes to the at-risk field.

2 · Testis

Structure

- Volume **15–25 mL**, length 4.5–5.1 cm. Capsule: visceral tunica vaginalis → **tunica albuginea** (smooth muscle) → tunica vasculosa.
- **Mediastinum testis**: posterior echogenic band on US. Appendix testis = **müllerian** remnant.
- ~250 lobules; **600–1200 seminiferous tubules** → rete testis. Interstitium **20–30%** of volume — **Leydig cells**. **Sertoli** tight junctions = blood–testis barrier.

Arterial supply — three sources

Artery	Origin	Point
Testicular (internal spermatic)	Aorta below renal artery	Main supply; larger than the other two combined
Deferential	Superior (or inferior) vesical	Vas, cauda; key collateral — lost after vasectomy
Cremasteric	Inferior epigastric	Tunica vaginalis; anastomoses at mediastinum

- Collaterals may save the testis after testicular artery ligation, but atrophy/azoospermia occur.

KEY POINT
Surgical rules: no traction suture through the lower pole; biopsy the **medial or lateral upper pole**; midportion has fewest vessels. After vasectomy the deferential collateral may be gone → preserve testicular artery in later cord surgery; counsel about devascularisation.

Veins — pampiniform plexus

Plexus surrounds the artery and cools it by **countercurrent heat exchange** to **2–4 °C below rectal**; varicocele disrupts this.

Group	Accompanies	Drains to
Anterior	Testicular artery	Inferior vena cava (IVC) (right) / left renal vein (left)
Middle	Vas	Pelvic veins
Posterior	Posterior cord	Epigastric and pudendal veins

- P1 Varicocele – Pathophysiology — *left vein into renal vein (**~90% of varicoceles left**), oblique right entry valve-like; middle/posterior groups cause recurrence*
- **High (retroperitoneal) ligation** takes testicular artery and vein above deferential/external vessels — collaterals suffice. **Inguinal/subinguinal** ligation takes veins only; arterial ligation here carries higher devascularisation risk.

Lymphatics and nerves

- Lymphatic division at varicocelectomy → **hydrocele**; microsurgery preserves lymphatics.
- Autonomic supply (renal/aortic plexuses; pelvic plexus via vas); **no somatic innervation**. Genital branch of genitofemoral supplies tunica vaginalis.
- **"Trifecta" nerve complex** (chronic orchialgia): perivasal, posterior periarterial/lipomatous, intracremasteric — targets of microsurgical cord denervation.

Blood–testis barrier

Sertoli–Sertoli tight junctions (basal vs adluminal compartments); also capillary endothelium and peritubular myoid cells → *Ch 67*.

Imaging

Modality	Role
Ultrasonography (US), 7.5–10 MHz + Doppler	**First line**: pain, mass, swelling, epididymo-orchitis, recurrent varicocele
Computed tomography (CT)	Not primary; cord/inguinal disease; air in Fournier gangrene
Magnetic resonance imaging (MRI)	Second line: indeterminate US, intra- vs extratesticular; nonpalpable testis (MR angiography)

- **Suspected torsion → explore**; Doppler must not delay surgery.
- Varicocele is a **clinical diagnosis**; US (erect, Valsalva) only if examination equivocal.

3 · Epididymis

- Caput, corpus, cauda on posterolateral testis. Epithelium: **principal** cells (blood–epididymis barrier), basal, **clear** cells (acidify lumen → quiescent storage). Cilia only in efferent ducts.
- Arteries: testicular → caput/corpus; deferential → cauda; cremasteric contributes.
- Nerves and smooth muscle increase toward the cauda (storage → propulsion).
- Inverted epididymis (caput at lower pole) in a minority.

Wolffian (mesonephric) duct	Müllerian (paramesonephric) duct
Epididymis, vas, seminal vesicle, ejaculatory duct, **appendix epididymis** (efferent ductules from mesonephric tubules)	**Appendix testis**, prostatic utricle

4 · Vas deferens

- Course: cauda → convoluted 2–3 cm → straight, **posterior** to cord vessels → inguinal canal → pelvic side wall → crosses **anterior to the ureter** → ampulla → joins seminal vesicle.
- Length **30–35 cm**; lumen **0.2–0.7 mm**; **three muscle layers** (longitudinal–circular–longitudinal). Lymph → external and internal iliac.
- ~Half the cord nerves surround the vas: sympathetic → emission peristalsis; sensory → chronic scrotal pain.
- **Vasogram**: hemivasotomy → 24 G angiocatheter → indigo carmine or contrast. Can cause obstruction → **only at reconstruction**.
- US confirms congenital absence of vas when examination equivocal.

5 · Seminal vesicles and ejaculatory ducts

- Behind bladder base; join ampulla → ejaculatory duct → verumontanum at **5 and 7 o'clock**.
- Bud from mesonephric duct at 12 weeks, linked to ureteric bud → **renal anomalies coexist**.
- Arteries: vesiculodeferential, inferior vesical; lymph → internal iliac. Noradrenergic → contraction at emission.
- Fluid ~70% of semen; alkaline; **fructose**, prostaglandins, **semenogelin** (coagulum, liquefied by PSA).
- **Transrectal ultrasound (TRUS)**: post-ejaculation AP diameter **>16 mm** or volume **>6 mL** = obstruction; guides aspiration/vesiculography. **MRI** modality of choice; CT for stones. Vesiculoscopy (4.5–6.9 Fr).

> **KEY POINT**
> A normal vesicle empties at ejaculation; one that stays large is the imaging signature of ejaculatory duct obstruction.

6 · Prostate

- P1 Anatomy of prostate and neurovascular bundles: clinical and surgical concepts in localised prostate cancer — *McNeal zones (peripheral **70%**, transition → BPH), fasciae, arteries, DVC, lymph, cavernous nerves*
- Normal **~25 mL**; transition zone grows fastest with age. Internal (preprostatic) sphincter → continence at rest. Peripheral zone hyperechoic on TRUS.
- "Capsule" 0.5–2 mm, deficient at apex.
- PSA normally luminal; disrupted architecture lets it reach serum. Cancer = **loss of basal cell layer**. Extraprostatic arterial anastomoses matter for embolisation.
- Parasympathetic → secretion; sympathetic (**α1**) → expulsion at emission.
- TRUS volume = L × W × H × **0.52** (underestimates). **Multiparametric MRI**: T2, diffusion, dynamic contrast; PI-RADS v2.1; targeted biopsy reduces insignificant-cancer detection.

7 · Urethra

- P1 Male urethra: embryology and anatomy, endoscopic stricture management, congenital anomalies and epispadias — *segments (membranous **2–2.5 cm**; Cowper 3/9 o'clock), epithelium, sphincter (Onuf), blood supply*
- Imaging: sonourethrography (anterior strictures), RUG (stricture, trauma; membranous narrowing normal), MRI (malignancy).

8 · Penis

Structure

- Paired **corpora cavernosa** (crura on pubic arch) + **corpus spongiosum** (bulb, urethra, glans).
- Septum incomplete distally → corpora act as **one unit** (†one-sided injection fills both).
- **Tunica albuginea** 2–3 mm: inner **circular** (intracavernosal struts), outer **longitudinal** (glans → inferior pubic ramus). Thin ventrally.
- **Buck fascia** encloses corpora, splits around spongiosum; contains (midline → lateral) **deep dorsal vein → arteries → nerves**.
- Sinusoidal smooth muscle **~45%** of volume; endothelial NOS sustains erection.

Arteries

Superficial: external pudendal (skin). Deep: internal pudendal → Alcock canal → **common penile artery**:

Branch	Supplies
Bulbourethral	Spongiosum, urethra, glans
Cavernosal	Enters at hilum, runs centrally → helicine arteries → sinusoids (erection)
Dorsal	Under Buck fascia → glans; circumflex branches

Accessory pudendal arteries may supplement or replace.

Veins, nerves

- Superficial dorsal vein (outside Buck) drains skin. **Deep dorsal vein** (under Buck) → periprostatic plexus.
- Distal two-thirds: sinusoids → subtunical plexus → **emissary veins** → circumflex → deep dorsal. Proximal third: cavernous/crural veins → internal pudendal.
- Shaft skin mobile; glans skin fixed to albuginea.
- **Dorsal nerve** (pudendal, **S2–S4**) — sensation. **Cavernous nerves** (pelvic plexus) — erection.

> **KEY POINT**
> Erection is veno-occlusive: filled sinusoids compress subtunical and emissary veins against the inelastic tunica; †tunical or smooth-muscle damage → venous leak.

Penile duplex (after intracavernosal injection)

Parameter	Threshold	Meaning
Peak systolic velocity	**<30 cm/s**	Arterial insufficiency
End-diastolic velocity	**>5 cm/s**	Veno-occlusive dysfunction
Resistive index	**<0.80**	Veno-occlusive dysfunction

MRI: fracture, prosthesis malposition, tumour.

67 Male Reproductive Physiology

1 · Hypothalamic–pituitary–gonadal axis

The HPG axis governs sex development, puberty, testosterone (endocrine testis) and sperm (exocrine testis). Pulse frequency and negative feedback set downstream responsiveness.

	Peptide hormones	Steroid hormones
Examples	LH, FSH	Testosterone, oestradiol
Storage	Secretory granules	Not stored — secretion = production
Receptor	Cell surface → cAMP	Intracellular → transcription; carried bound in plasma

Hypothalamus

- GnRH: decapeptide (preoptic, arcuate nuclei) → portal system; sole function LH/FSH release. Half-life **5–7 min**.
- Rhythms: pulsatile (every **90–120 min**), circadian (testosterone highest early morning), seasonal.
- **Pulsatility essential**: continuous GnRH agonist (leuprolide) → desensitisation → castrate testosterone.
- **Kallmann syndrome**: GnRH neurones fail to migrate → hypogonadotrophic hypogonadism. †Anosmia — neurones migrate with the olfactory placode.

GnRH modulation	Factors
Negative	Opioids (β-endorphin), FSH, LH, testosterone
Positive	Prostaglandin E2, insulin, **kisspeptin** (puberty), **leptin**
Variable	Dopamine (catecholamines)

Anterior pituitary

- LH/FSH glycoproteins: common **α subunit** (shared with TSH, hCG) + specific **β subunit**; act via cAMP.
- **Prolactin** may raise Leydig LH receptors and support libido. **Hyperprolactinaemia abolishes gonadotrophin pulsatility** → low testosterone. Low prolactin not pathological.

	LH	FSH
Pulses	8–16/24 h; follow GnRH	Smaller; partly GnRH-independent (inhibin, activin)
Target	**Leydig cell**	**Sertoli cell** (and spermatogonia)
Action	Steroidogenesis: cholesterol → pregnenolone → testosterone	Tubule growth; **initiates spermatogenesis at puberty**; maintains quantitative output
Main feedback	Testosterone (hypothalamus)	Oestradiol (pituitary) + **inhibin B**

Testicular hormones and feedback

- Testosterone ~**5 mg/day**, pulsatile. **DHT** via 5α-reductase — more potent, needed in most peripheral tissues (not testis/muscle); **oestradiol** via aromatase — feedback especially on FSH.
- FSH induces Sertoli ABP, transferrin, lactate, inhibin, growth factors.
- **Inhibin** (Sertoli, 32 kDa) suppresses FSH; **activin** (TGF-β family) stimulates FSH.

KEY POINT

Two loops: LH ↔ Leydig (testosterone) and FSH ↔ Sertoli (inhibin, oestradiol). **LH reports Leydig function; FSH and inhibin B report the tubules** — high FSH with low inhibin B = spermatogenic failure.

Development and ageing

- **SRY** (Yp) encodes an HMG-box protein that bends DNA → testis. **DAX1**, **WNT4** = "antitestis" genes that can induce female development despite SRY — female pathway not a passive default.

Fetal cell	Product	Effect
Leydig	Testosterone	Wolffian duct → internal genitalia
Leydig	Insulin-like peptide 3 (INSL3)	Transabdominal testicular descent
Target tissue	DHT	External genitalia
Sertoli	MIS/AMH	Müllerian regression; keeps germ cells out of meiosis

- Neonatal surge → childhood quiescence → puberty (~12 years), triggered by kisspeptin, leptin, melatonin.
- **Ageing**: testosterone ~one-third lower by 7th decade; FSH ~threefold; fewer Leydig cells, more SHBG, lost diurnal rhythm, blunted feedback — peripheral and central failure.

2 · Testis

Architecture and blood flow

- **600–1200 tubules**; ~**700 million Leydig cells** in young adults.
- Cord holds one artery in ~half, two in ~30%, three in ~20% — relevant to microsurgical varicocelectomy; single artery ligation can cause atrophy.
- **Countercurrent exchange** cools arterial blood **2–4 °C** below core and recycles testosterone vein → artery; lost in varicocele, cryptorchidism.
- Tubular fluid reabsorbed in rete/efferent ductules under oestrogen control.

Leydig cells and testosterone synthesis

- Differentiate by week 7 (hCG/LH); neonatal wave ~2–3 months imprints hypothalamus, liver, prostate.
- **Rate-limiting step**: LH → cAMP → **StAR protein** moves cholesterol into inner mitochondrial membrane → pregnenolone → smooth ER enzymes (3β-HSD, 17α-hydroxylase/17,20-lyase, 17β-HSD) → testosterone. Mutations → 46,XY androgen biosynthesis disorders.
- FSH, prolactin enhance LH response; oestrogens, androgens inhibit.
- †Circulating testosterone: ~44% SHBG (tight), ~54% albumin (loose), ~2% free; free + albumin = bioavailable.

Testosterone peak	Timing	Role
Fetal	12–18 weeks	Builds reproductive tract
Neonatal	~2 months	Imprints target tissues
Pubertal	12–17 years	Masculinisation
Adult	Peak 2nd–3rd decade → decline	Maintains androgen-dependent organs

Sertoli cells and blood-testis barrier

- P3 P1 Spermatogenesis; Male reproductive immunology — *Sertoli nurse functions, ABP (intratubular androgen ~**50-fold** serum), basal/adluminal compartments, anatomical + functional barrier, no antibodies before puberty, chemotherapy sanctuary*
- Sertoli tight junctions strongest in body; open/close to let spermatocytes pass. **Inhibin B** = marker of Sertoli function. c-kit/kit ligand signalling; spermiation.
- Peritubular myoid cells: contractile; secrete P-Mod-S modulating Sertoli function.
- TGF-β3 and TNF-α reversibly open the barrier (reduce occludin, ZO-1).

3 · Spermatogenesis

- P3 Spermatogenesis — ***~64 days*** *(42–76); proliferation (Ad, Ap every 16 days, B) → meiosis → spermiogenesis (acrosome from Golgi, flagellum from centriole) → spermiation; six stages, helical wave; cytoplasmic bridges*
- Output ~**123 million/day** (~1000 per heartbeat). †Needs high intratesticular testosterone → exogenous testosterone suppresses spermatogenesis.
- Primary spermatocytes: preleptotene → leptotene → zygotene → pachytene. Spermatids Sa → Sb → Sc → Sd1 → Sd2.
- **Cycle** = succession of stages at one point; **stage** = cell association; **wave** = arrangement along tubule.

Stem cells

- Primordial germ cells → gonadal ridge → gonocytes → spermatogonia. Failed migration → extragonadal germ cell tumour, Sertoli cell–only.
- Self-renewal via kit ligand/c-kit and GDNF.
- "Selfish spermatogonia" (FGFR2/FGFR3 mutations) → paternal-age disorders, spermatocytic seminoma.

Meiosis

	Mitosis	Meiosis
Divisions / daughters	1 / 2 identical	2 / 4 varied, haploid
Synapsis, crossover	None	Prophase I (synaptonemal complex, chiasmata)
Centromere division	Anaphase	**Anaphase II**

- **Defective recombination** in ~half of maturation arrest; altered crossover → aneuploidy → more chromosomal abnormalities in IVF/ICSI offspring.

Genetics

Y microdeletions (AZF): **15%** of azoospermic and 6–8% of severely oligospermic men — commonest defined molecular cause.

Deletion	Histology	Sperm retrieval
Complete AZFa	Sertoli cell–only	Highly unlikely
Complete AZFb	Maturation arrest	Highly unlikely
AZFc	Hypospermatogenesis / focal spermatogenesis	**Often found**
Partial AZFa/b	—	Ejaculated sperm possible

X-linked: SOX3, TEX11. **Paternal age**: ~23 spermatogonial divisions/year → de novo dominant mutations (achondroplasia, Apert, Marfan), autism, schizophrenia, aneuploidy, rising **DNA**

fragmentation → poor embryos, miscarriage.

4 · Epididymis

- 3–4 m duct: caput (8–12 efferent ductules) → corpus → cauda. Testicular sperm **immotile, cannot fertilise** (except ICSI).
- Principal cells (stereocilia = microvilli), basal cells; ciliated cells only in efferent ductules. Barrier less tight than testis. Muscle and sympathetic nerves increase toward cauda.

Functions

- **Transport**: **2–12 days**, mainly spontaneous rhythmic contraction. Recent ejaculation shortens only cauda transit.
- **Storage**: cauda holds ~half of 155–209 million sperm per side. Prolonged storage → more DNA fragmentation.
- **Maturation**: progressive motility, zona binding/penetration; surface charge, disulfide cross-linking, membrane remodelling. **Fertility reached mostly by late corpus/early cauda**.
- Epididymosomes deliver small RNAs to the embryo.

> **KEY POINT**
> Pregnancy occurs even after vasoepididymostomy at the efferent ductules, but **the more distal the anastomosis, the better the fertility**. Vasectomy >15 years → lower motility after reversal (epididymal dysfunction).

- **Androgen dependent** (higher levels, mainly DHT) → impaired semen with 5α-reductase inhibitors.
- **Heat** abolishes storage; **denervation** (spinal cord injury, diabetes) impairs transport.

5 · Vas deferens, seminal vesicles, ejaculatory ducts

- Vas: wolffian, 30–35 cm; **muscle:lumen ~10:1** — most muscular hollow viscus. Lumen 200–700 µm.
- Adrenergic (hypogastric) + purinergic innervation → peristalsis before emission. Testicular-side segment has fewer nerves. Vasal reserve ~130 million sperm.
- **Seminal vesicle**: 5–7 cm coiled tubule; ~**80%** of seminal volume (late fractions); alkaline, **fructose**, prostaglandins; **semenogelin I** → coagulum. Antioxidants: glutathione peroxidase, SOD, catalase.
- **Ejaculatory duct**: no sphincter — reflux prevented by **acute insertion angle**.
- Vesicle–duct unit like bladder–urethra: obstruction **physical** or **functional** (vesicle myopathy: diabetes, drugs); static TRUS may not distinguish.
- **Ejaculate pH <7.2, low volume, no fructose** → absent/blocked seminal vesicles (CBAVD, ejaculatory duct obstruction).

6 · Spermatozoon, capacitation and fertilisation

Part	Contents
Head (4.5 × 3 µm)	Compacted nucleus; acrosome (enzymes)
Neck	Proximal centriole; axoneme starts
Midpiece	~75 mitochondria in helix, outer dense fibres; ends at annulus
Principal piece / endpiece	Axoneme with fibrous sheath

- Length ~60 µm. Axoneme **"9 + 2"**; **dynein** (Mg^{2+}-ATPase) slides doublets. Inner dynein arm mutants: no motility; outer arm: reduced.
- Sperm mtDNA mutates 40–100× faster; **paternal mitochondria degraded** in egg (ubiquitin) → maternal inheritance.
- **Capacitation** in female tract: hyperactivated motility, readiness for **acrosome reaction**; coagulum prevents premature capacitation.
- Binding zona **ZP3** → acrosome reaction; **fertilin (PH30)** → oolemma fusion; sperm **PLCζ** → calcium oscillations = oocyte activation.

Sperm defects

- Strict (Kruger) morphology; head defects rarely reflect DNA quality, except **pinhead** and **multiple-tailed**.
- **Globozoospermia** (round head, no acrosome): fertilisation fails even with ICSI — PLCζ deficiency.

	Ciliary dyskinesia (immotile cilia)	MMAF (fibrous sheath dysplasia)
Motility	<10%	Severely reduced
Morphology	Normal on light microscopy; axonemal defect on EM	Short, bent, coiled, absent flagella
Cause	Recessive; respiratory disease, dextrocardia (Kartagener)	Genetic; also varicocele, ROS
Treatment	IVF-ICSI	Correct cause; else IVF-ICSI

7 · Ejaculation physiology †

ALGORITHM

Emission (sympathetic, T10–L2, hypogastric; α1): vas, seminal vesicle, prostate contract → posterior urethra; **bladder neck closes** → **Expulsion** (somatic, S2–S4 pudendal): rhythmic bulbospongiosus/pelvic floor contractions, relaxed external sphincter → antegrade ejaculate.

- Fractions: sperm-rich and prostatic first, seminal vesicle last.
- P3 Retrograde ejaculation — *failed bladder neck closure (α-blocker, diabetes, bladder neck surgery, RPLND) → retrograde; loss of emission → anejaculation*

68 Integrated Men's Health: Androgen Deficiency, Cardiovascular Risk and Metabolic Syndrome

1 · The men's health centre

- Multidisciplinary (urology + primary care, cardiology, gastroenterology). Men avoid preventive care but seek help "below the belt" (ED, infertility, voiding) → urologist is the **front door** who triages the whole man.
- Advanced practice providers and telemedicine extend reach; shared decision making.

Age band	Dominant need	Key partners
18–39	Sexual, reproductive, testicular ("rule out hypogonadism"); preventive	Sports medicine, physiotherapy, mental health
40–64	Prostate and colon screening; disease management	GI, cardiology, endocrinology, bariatrics, pulmonology
65+	Falls, neurodegeneration, hearing/vision, nutrition	Geriatrics, neurology, audiology, ophthalmology

2 · The longevity gap

- Life expectancy: males **76.1** vs females 81.1 years (2016).
- Males: higher mortality in **12 of 15** leading causes; largest gap = **suicide (>3-fold)**. More early MI; T2DM rising faster.
- Causes: hormonal (testosterone harm, oestrogen cardioprotection), behavioural, risk-taking.
- Prostate cancer: only US malignancy with rising stage IV at diagnosis → discuss screening.

3 · Urologic screening

- Screen sexual and voiding complaints together; PSA and urinalysis per guidelines.

Questionnaire	Screens
SHIM	Erectile function
ADAM	Androgen-deficiency symptoms
IPSS	LUTS

4 · Urologic disorders of focus

Testosterone deficiency (TD)

- P4 Male menopause (andropause, late-onset hypogonadism) — *low T on more than one occasion + symptoms; AUA **<300 ng/dL** on 2 morning samples; LH separates primary/secondary, prolactin if LH low; TRT contraindications; monitor T, haematocrit (**>54%** hold), PSA at 3–6 months then yearly*
- Haematocrit: AUA withholds at >50%, Endocrine Society at >54% → reduce dose/phlebotomy. PSA rise or abnormal DRE → urology.
- T falls with age from testicular and hypothalamic–pituitary defects; symptoms overlap with comorbidity. Exclude secondary causes before TRT.
- Chronic opioids (~5 million US men) → Endocrine Society: screen, consider TRT.

- **Randomised TRT benefit**: obesity, metabolic syndrome, diabetes, muscle, bone. Mood, cognition, sexual benefits observational.
- T and depression share causes and symptoms, but **TRT is not a treatment for depression**; likeliest responders: clearly deficient, <60 y, milder depression.

> **KEY POINT**
> † **Fertility caution**: exogenous T suppresses LH/FSH and intratesticular T → oligo/azoospermia. Men wanting children: avoid TRT; use clomiphene, hCG or aromatase inhibitors → *Ch 69*.

† TRT	Points
Contraindications	Near-term fertility; breast cancer; untreated prostate cancer; haematocrit >50%; untreated severe OSA; uncontrolled heart failure; MI/stroke within 3–6 months
Target	Mid-normal T (≈450–600 ng/dL)
Formulations	Gels, injections, pellets — each with its own monitoring plan

Erectile dysfunction

- >50% of men aged 40–70; multifactorial.
- **Every man with ED, of any age, is at CVD risk**. †Small penile arteries → ED precedes coronary events by ~2–5 years.
- Lifestyle → PDE5 inhibitors (never with nitrates) → intracavernosal injection, intraurethral, vacuum → prosthesis (highest satisfaction; counsel irreversibility, infection, malfunction).
- Platelet-rich plasma, low-intensity shock wave: trials only.

BPH and infertility

- BPH: mostly medical; refer when medication fails, not tolerated or not wanted.
- Infertility: no pregnancy after 1 year; male factor in ~half; semen analysis is cornerstone → *Ch 69*.

5 · Cardiovascular prevention

- ACC/AHA ASCVD calculator → 10-year risk; ignores family history and diabetes control.
- Only **LDL-C lowering** benefits primary prevention (not raising HDL or lowering triglycerides). Lipid screening from 20.

Statin decision (ACC/AHA)	Action
LDL-C ≥190 mg/dL	High-intensity statin ± nonstatin
LDL-C 70–189, 10-year risk ≥ **7.5%**	Moderate–high intensity statin
Risk 5–7.5%	Moderate intensity may be started
Statin-intolerant, risk >20%	Ezetimibe or PCSK9 inhibitor

- Moderate intensity preferred in primary prevention. USPSTF/NICE/ESC start at ≥10% risk.
- **CAC score**: asymptomatic men >40, most useful at **intermediate risk**; zero → <1% atherosclerosis (less reliable in diabetics).
- Lipoprotein(a): near-linear risk, unaffected by statins.

BP (ACC/AHA 2017)	Systolic	Diastolic
Normal	<120	and <80

BP (ACC/AHA 2017)	Systolic	Diastolic
Elevated	120–129	and <80
Stage 1	130–139	or 80–89
Stage 2	≥140	or ≥90

Hypertension = most important modifiable risk. Elevated/stage 1 → lifestyle (salt, weight, DASH, exercise, alcohol). Drugs: stage 2, or stage 1 + age ≥65, T2DM, CVD, CKD or 10-year risk ≥10%.

6 · Obesity and metabolic syndrome

- BMI: overweight 25–29.9; obesity I 30–34.9, II 35–39.9, III ≥40.
- Weight gain ≥5 kg after 20 raises coronary and T2DM risk regardless of BMI.
- BMI ≥30, or 25–29.9 with CVD risk/OSA/osteoarthritis → weight loss; GLP-1 agonists (semaglutide) FDA-approved.
- †Metabolic syndrome (ATP III): ≥3 of waist >102 cm, TG ≥150 mg/dL, HDL <40 mg/dL, BP ≥130/85, glucose ≥100 mg/dL. Low T ↔ metabolic syndrome bidirectional (adipose aromatase ↑ oestradiol → ↓ LH).

7 · GI, skin and mental health

- **Constipation**: evaluate in every man with UTI or retention. Alarm features → colonoscopy.
- Iron deficiency anaemia: >30-fold GI malignancy risk. Colorectal screening **45–75** years. Long-term PPI → check for Barrett.
- Skin: basal cell commonest; melanoma (ABCDE). Psoriasis linked to MI. Alopecia: 5α-reductase inhibitors + minoxidil.
- Suicide much higher in men; substance use, autism, ADHD male-predominant.

69 Male Infertility

1 · Definition and overview

- Infertility: no pregnancy after **1 year** unprotected intercourse; ~15% of couples; male and female contribute equally — always evaluate the man. Most conceptions in first 3–6 cycles → may evaluate after 6 months.
- Reproduction is **probabilistic**: tests modify a probability, not a verdict.
- Diagnoses: idiopathic (largest), **varicocele (commonest identifiable)**, obstruction; others uncommon.

† Classification	Examples
Pre-testicular (endocrine)	Hypogonadotropic hypogonadism (HH), Kallmann, prolactinoma, exogenous androgens, opioids, obesity, CAH
Testicular	Klinefelter, Y microdeletion, cryptorchidism, torsion, orchitis, varicocele, chemo/radiation, Leydig failure
Post-testicular	Obstruction (epididymal, vasal, ejaculatory duct), CBAVD, ejaculatory dysfunction, antisperm antibodies, structural sperm defects

2 · History

- P2 Discuss the role of urologist in reproductive endocrinology and male infertility practice — *history, examination, endocrine tests (LH localises, FSH/inhibin B = germ-cell mass, prolactin), HH/hyperprolactinaemia/CAH treatment, surgical roles, ART choice, testis cancer in azoospermia*
- **Female partner's age = most important determinant**; fecundity falls after 35.
- Paternal age ≥40: counsel on offspring bipolar disorder, schizophrenia, autism, ADHD (ASRM/AUA 2021). Prior paternity excludes many congenital causes.
- Semen peaks after **1–2 days' abstinence**; advise daily intercourse around ovulation.
- Lubricants (K-Y, Astroglide, Replens, saliva) impair sperm; **Pre-Seed** safe.
- Mnemonic **TICS**: Toxins, Infection, Childhood history, Sexual history.
- Cancer impairs spermatogenesis even before chemo. Mesh herniorrhaphy can entrap the vas.

Gonadotoxins and drugs

> KEY POINT
> **Exogenous testosterone (T) is a contraceptive**: aromatised to E2 → pituitary feedback → ↓ LH → ↓ intratesticular T → ↓ spermatogenesis. Recovery usually ≥4 months after stopping (≤2 years' use); after longer use, unknown.

- Antiandrogens, spironolactone, corticosteroids, oestrogens, anabolic steroids, some antiretrovirals; 5ARIs (limited).
- Cannabis: dose-dependent ↓ T and count. Alcohol: ↑ aromatisation. Smoking: oxidative stress, DNA damage; additive with varicocele. Maternal smoking → sons with smaller testes.

Infection and childhood conditions

- Infection: **anatomical** (duct scarring; TB) or **functional** (leucocytes, ROS, cytokines) injury.
- Torsion: contralateral biopsy usually abnormal; ~half have later spermatogenic effects; antisperm antibodies.

Cryptorchidism (after orchidopexy)	Unilateral	Bilateral
Paternity	Near normal	Markedly reduced
Inhibin B	~⅔ of controls	~⅓ of controls

- **Laterality is the key prognostic factor.** Germ-cell/Sertoli lesion, begins fetally, worsens with time.
- Orchidopexy within **18 months** (AUA); germ cells lost from 15–18 months. If missed, surgery before 10 years improves later retrieval.

3 · Physical examination

- Eunuchoid: arm span >5 cm over height. Klinefelter habitus often absent. Examine standing.
- Engorged epididymis → obstruction. Testis size reflects spermatogenic mass.

Testis size	Abnormal
Seager caliper (long axis)	**≤4.6 cm**
Prader orchidometer (volume)	≤20 mL
Scrotal ultrasound	Most accurate; added value unclear

- **Vas**: slips from the fingers three times → absent (Meacham). Unilateral absence → renal US. Bilateral → CFTR + renal US.
- Palpable seminal vesicles → EDO.

Varicocele grade	Definition
Subclinical	Only on Doppler — **do not treat**
I	Palpable only with Valsalva, standing
II	Palpable without Valsalva, not visible
III	Visible through scrotal skin

4 · Endocrine evaluation

> **ALGORITHM**
> Morning: total T + SHBG + albumin → calculated bioavailable T → LH, FSH → E2; prolactin only if indicated

- Adequate: total T ≥ **280–300 ng/dL**. Low T common even with density >20 million/mL → assess every infertile man (not needed above ~66 million/mL).
- Bioavailable = free (0.5–3%) + albumin-bound (54–68%); SHBG-bound (30–44%) unavailable; Vermeulen formula.
- SHBG ↓ (obesity, hypothyroidism, diabetes, steroids) → total T under-reads; ↑ (ageing, cirrhosis, hyperthyroidism, HIV, anticonvulsants) → over-reads.
- Borderline: 3 samples 20 min apart, averaged.
- FSH: abnormal concentration odds rise from 4.5 IU/L. AMH experimental.
- T (ng/dL) : E2 (pg/mL) **<10:1** → aromatase excess.
- Prolactin 20–50 µg/L: no further work-up; significant rise → pituitary MRI.

5 · Semen analysis

- P4 Semen analysis nomenclature — *collection, ≥2 samples 2–3 weeks apart; WHO 2021 5th centiles (volume **1.4** mL, **16** million/mL, 39 million, 42/30%, **4**% strict forms, 54% vitality); terms; only azoospermia definitive; HOS test*
- WHO 6th 50th centiles: 3.0 mL, 66 million/mL, 210 million, 64% total, 55% progressive, 14% forms, 78% vitality. Reference, not pass/fail; poor discriminator (AUC ≈0.6).
- Legacy WHO concentration threshold 20 million/mL; Guzick subfertile/fertile 13.5/48 million/mL.
- Brown semen often SCI. Isolated teratozoospermia does not predict natural, IUI or IVF outcome.
- **Low volume (<1.0 mL)**: EDO, CBAVD, hypoandrogenism, retrograde ejaculation, neuropathy, α-blockers → **postejaculatory urinalysis** (significant if urinary ≈ antegrade) → TRUS if none.
- †Sperm DNA fragmentation (SCSA, TUNEL): not routine; recurrent pregnancy loss or failed ART; raised by varicocele, smoking, oxidative stress.

6 · Genetic testing

> **ALGORITHM**
> NOA or density **<5 million/mL** → karyotype + Y-microdeletion (ASRM/AUA 2021) · CBAVD (or idiopathic EDO) → CFTR testing of man **and partner**

- P3 Genetic causes of male infertility. Enumerate and classify. — *AZFa (USP9Y, DDX3Y)/AZFb complete deletion → no retrieval; AZFc (DAZ) → sperm often found, sons inherit; CFTR; Kallmann; androgen receptor; globozoospermia, AURKC, PCD*
- AZFb genes EIF1AY, KDM5D. Yq palindromic amplicons make region fragile.
- AZFc with oligozoospermia → progressive decline → **cryopreserve sperm**.

7 · Imaging

- Not for screening (incidental microlithiasis harms). Varicocele is clinical.
- P3 Ejacultory duct obstruction — *TRUS vesicle AP **>1.5 cm** ± midline cyst (low specificity); aspiration ≥3 sperm/HPF; vesiculography, chromotubation; <5% of azoospermia; idiopathic EDO CFTR in ~half; TURED with TRUS guidance*

8 · Azoospermia: obstructive (OA) vs non-obstructive (NOA)

	OA	NOA (spermatogenic dysfunction)
FSH	**≤7.6 IU/L**	>7.6 IU/L
Testis long axis	>4.6 cm	≤4.6 cm
Management	Reconstruction → *Ch 70* or sperm retrieval	Genetic tests; micro-TESE + ICSI

> **ALGORITHM**
> Azoospermia → vasa absent? Yes → CFTR (both partners) · No → volume <1 mL? Yes → postejaculatory urinalysis: sperm → retrograde ejaculation; none → TRUS for EDO · Normal volume → FSH + testis size → OA vs NOA

- NOA histology: hypospermatogenesis; maturation arrest; Sertoli cell–only.

9 · Specific causes and treatment

Klinefelter syndrome (47,XXY)

- P4 Klinefelter's syndrome. — *commonest genetic cause; 1 in 500–1000; non-mosaic 80–90%; small firm testes, ↑ FSH/LH, low T; ejaculated sperm ~8%; micro-TESE ~**50%**; mediastinal GCT*
- **Trap**: exogenous T suppresses residual spermatogenesis → extract sperm before T; use aromatase inhibitors/hCG if fertility wanted.

Hypogonadotropic hypogonadism — most treatable cause

State	Hormones	Treatment
Kallmann (anosmia; KAL1, GNRHR)/complete HH	↓ LH, ↓ FSH	hCG ± rFSH or hMG
Isolated LH deficiency ("fertile eunuch")	↓ LH, normal FSH	hCG
Isolated FSH deficiency	↓ FSH	FSH
Hypothalamic GnRH deficiency	↓ GnRH	Pulsatile subcutaneous GnRH every 2 h, ≥6 months
Partial HH (common)	LH inappropriately low for T	SERM or aromatase inhibitor

- hCG 1500–5000 IU 2–3×/week (max 10,000/week), titrate to T; add hMG/rFSH 75 IU 2–3×/week; 1–2 years after androgen abuse. Prolactinoma: cabergoline. CAH: corticosteroids; watch testicular adrenal rest tumours.

Empirical medical therapy

Drug	Dose	Notes
Clomiphene (SERM)	25 mg daily or 50 mg alternate days; max 100 mg	Blocks E2 feedback → ↑ LH/FSH; off-label; no PSA/haematocrit change; target 600–800 ng/dL
Anastrozole / letrozole	Anastrozole 1 mg daily	↑ T, ↓ E2; when T:E2 <10:1 (obesity); caution bone density

Primary failure and androgen resistance

- ↑ LH, ↓ T: no accepted therapy; exogenous T contraindicated → sperm extraction.
- Androgen receptor defect (Xq11–12, long CAG): ↑ T, ↑ LH, ↑ E2 → ICSI. 5α-reductase type 2 deficiency: ICSI pregnancies reported.

CBAVD

- P3 CBAVD — *genital form of CF; mild CFTR alleles; normal spermatogenesis, low-volume azoospermia; partner carrier ~1 in 20; renal agenesis ~1 in 10; retrieval + ICSI*

Varicocele

- P3 Varicocele: controversies and present status of treatment — *prevalence, left predominance, right-sided → exclude tumour; heat/ROS/DNA injury; repair palpable + abnormal semen, never subclinical; may down-stage ART*
- ~15–20% of all men vs 35–50% of infertile men → *Ch 70*.

Antisperm antibodies and structural defects

- Antisperm antibodies: washing fails, steroid trials conflict → **ICSI**.
- Globozoospermia: ICSI ± oocyte activation. Macrocephalic multiflagellate (AURKC): ICSI not recommended. PCD: ICSI (testicular sperm better).

10 · Lifestyle and obesity

- Obesity: ↑ aromatase → ↑ E2 → ↓ LH/T; ↓ SHBG, ↓ inhibin B; hormone-independent DNA effects → hormones alone may not restore fertility. Worsens ART outcomes; weight loss advised.
- Stop smoking, cannabis, excess alcohol, anabolic steroids; avoid heat; psychological support.

11 · Assisted reproduction

- Preparation: density gradient, swim-up. ICSI needs one sperm per oocyte, bypasses selection.
- †By total motile sperm count: IUI if processed TMSC ≳5–10 million; IVF below or after failed IUI; ICSI for severe oligozoospermia, retrieved sperm, antisperm antibodies, prior fertilisation failure.
- †Retrieval: OA — PESA/MESA or testicular aspiration; NOA — micro-TESE.

70 Surgical Management of Male Infertility

1 · Principles

- IVF/ICSI treats even NOA but is costly, burdens the female partner and bypasses natural barriers (genetic transmission, birth defects).
- Treating the man (reconstruction for OA, varicocelectomy) is safest and most cost-effective and moves the couple **down the ladder**: donor sperm → IVF/ICSI with testicular sperm → IVF/ICSI with ejaculated sperm or IUI → natural conception.

2 · Surgical anatomy

Structure	Arteries (origin)
Testis	**Testicular** (aorta) — main; deferential (internal iliac/superior vesical); cremasteric (inferior epigastric; mainly tunica vaginalis)
Epididymis	Superior/medial epididymal (testicular a.) → caput; inferior epididymal (deferential a.) → cauda; interconnected
Vas	Abdominal end: deferential only. Testicular end: deferential + inferior epididymal. **No supply from cremasteric or cord vessels**

- Testicular artery **exceeds deferential + cremasteric combined**. After vasectomy the vasal artery is presumed lost → preserve testicular artery at later surgery.
- Epididymis can be dissected off the testis (inferior/medial epididymal arteries ligated) if the **superior epididymal artery is intact** → gains **4–6 cm**.
- Vas blocked at two sites with vessels interrupted at both → intervening segment fibroses → **never two simultaneous vasovasostomies on one vas**; excise, do one tension-free anastomosis.
- **Excurrent ducts**: rete → 7–11 efferent ducts → caput → **single tubule** from caput–corpus junction → vas. Injury distal to caput (even **one stitch**) obstructs the whole side; single puncture safe only at most proximal caput.

3 · Diagnostic testis biopsy

- P1 Testicular biopsy and TESA (testicular sperm aspiration) — *indications (azoospermia, normal testes, palpable vasa, normal FSH); diagnostic + therapeutic; open = gold standard; never formalin; percutaneous 14 G / 23 G*

> **ALGORITHM**
> Cord block → larger testis (contralateral to varicocele) → 2 cm incision → open tunica vaginalis → stay sutures → 3–4 mm albugineal incision, bipolar crossing vessels → pea-sized sample, iris scissors (no forceps) → touch prep → wet prep → negative: other side → close tunica vaginalis continuous 5-0; no antibiotics

- Microscope 10–25× → sample larger tubules; non-absorbable suture marks a positive site for future TESE.

Complication	Point
Haematoma — most common	Prevent: magnification, bipolar, continuous tunica vaginalis closure
Epididymal biopsy — most serious	Sperm in that tubule → obstruction now certain

4 · Vasography

- **Absolute indications (all three)**: azoospermia + complete spermatogenesis on biopsy + ≥1 palpable vas. Relative: severe oligospermia with normal biopsy, high sperm-bound antibodies, low volume with poor motility (partial EDO).
- **Only at planned reconstruction** (scar; stricture). Cannot diagnose epididymal obstruction — commonest non-vasectomy site.

> **ALGORITHM**
> High scrotal incision → vas at straight–convoluted junction → hemivasotomy at 25× → **vasal fluid examined first** → motile sperm: barbotage, cryopreserve **before any dye** → 24-G sheath into abdominal end, 1 mL saline: easy flow = patent → (rarely) dilute indigo carmine, blue urine confirms → formal contrast only for obstruction above internal ring → close 10-0 mucosa, 8-0 muscularis

Vasal fluid / vasogram	Meaning → action
No sperm, abdominal end patent	Epididymal obstruction → vasoepididymostomy
Copious fluid, many sperm	Vasal obstruction or EDO → locate (vasogram, 2-0 Prolene threaded)
Block at ejaculatory ducts; both vasa fill from one side	EDO / midline cyst → TURED
Vas ends blindly	Partial vasal agenesis → CFTR testing; retrieve + freeze
Inguinal block	Inguinal or crossed vasovasostomy
Many sperm, normal vasogram	Retrograde ejaculation, emission failure, aperistalsis

- **Indigo carmine, not methylene blue** (kills sperm; only after banking).
- Complications: **stricture** (repeated punctures, poor closure, non-water-soluble contrast), devascularisation, haematoma, granuloma.
- **Transrectal seminal vesiculography**: TRUS aspiration of dilated SVs; sperm = ≥1 vas/epididymis patent → contrast + indigo carmine → TURED without scrotal exploration. No sperm → secondary epididymal obstruction → retrieval + ICSI.

5 · Vasovasostomy (vasectomy reversal)

Evaluation

- Prior fertility proves spermatogenesis; FSH if testes small/soft (raised = poorer).
- Indurated epididymis → vasoepididymostomy likely; **sperm granuloma at testicular end → better prognosis** (vents pressure) regardless of interval; large gap → inguinal extension; groin scars → iatrogenic obstruction.
- Whole sperm in centrifuged pellet → sperm certain in ≥1 vas. Low volume → TRUS.
- General anaesthesia preferred; operate only if able to do vasoepididymostomy.

Approach and preparation

- Scrotal for most; inguinal if hernia/orchidopexy obstruction; robotic for transection above inguinal ligament.
- **Never strip the perivasal sheath**; excise granuloma staying on the vas.
- **Length**, escalating: dissect to external then internal ring → free convoluted vas (4–6 cm; **do not unravel**) → reroute beneath inguinal floor → epididymis off testis → gaps up to **10 cm**.

- Cut through slotted clamp. Healthy end: white mucosal ring, soft muscularis, bleeding edge. **Gritty muscularis or poor bleeding → recut.**
- Testicular end: **aspirate, never flush**. Abdominal end: flush saline to prove patency.

Intraoperative fluid grading — the decision

Vasal fluid	Microscopy	Operation
Copious, crystal clear, watery	No sperm	Vasovasostomy
Copious, cloudy, thin	Sperm with tails	Vasovasostomy
Copious, creamy yellow	Many sperm heads	Vasovasostomy
Scant, granuloma present	Barbotage: sperm	Vasovasostomy
Thick white **toothpaste**	No sperm/heads	**Vasoepididymostomy**
Scant white thin, or dry without granuloma	No sperm/heads	**Vasoepididymostomy**
Any	Grape-like clusters of heads	**Vasoepididymostomy**

> **KEY POINT**
> Clear watery fluid still gets vasovasostomy: epididymis not yet blown out; sperm usually return. Toothpaste fluid = secondary epididymal blow-out → open tunica vaginalis; discrete level (dilated above, collapsed below) confirms. When in doubt or inexperienced, do vasovasostomy.

Anastomotic principles

Principle	Reason
Accurate mucosa-to-mucosa	Testicular lumen 2–5× abdominal
Leakproof	Sperm antigenic; fluid lacks clotting factors — seal depends on mucosal sutures
Tension-free	Tension → early sperm then falling counts → azoospermia
Good blood supply; healthy mucosa/muscularis; atraumatic	Recut until bleeding and soft; redo if back-walled or torn

- **Multilayer microdot**: six mapped dots → **six 10-0 mucosal sutures** (double-armed, inside-out) → 9-0 deep muscularis → 9-0 adventitia → 8-0 sheath. Alternative two-layer: 10-0 mucosa + 8-0 full thickness.
- **ReVas** (reinforcing vasal): sheath 3 cm either side looped with 5-0 Prolene + 5–10 reinforcing sutures → tension relief, higher patency.
- **Convoluted vas**: transverse cut, do not unravel, small bites, reinforce.
- **Crossed vasovasostomy**: unilateral inguinal obstruction with contralateral atrophic testis, or one side unreconstructable; testicular transposition if vas very short.
- **Varicocele + reversal**: vasal vessels compromised → stage varicocelectomy ≥6 months later.

Post-operative care and outcomes

- Scrotal support, no heavy lifting 4 weeks; ejaculation 2 weeks, intercourse 4 weeks.
- Semen at **6 weeks**, then 6 months if no pregnancy. Poor motility → 4-week prednisone. Azoospermia at 6 months → redo.
- Perivasal haematoma (commonest); anastomotic granuloma (harbinger of obstruction); **late stricture** — falling motility then count. **Bank sperm once motile sperm appear.**

- Sperm in vasal fluid ≥1 side → sperm return in virtually all; pregnancy ~half (vasectomy interval, female age).

6 · Vasoepididymostomy

- Single tubule ~200 µm, 12–15 feet; wall thinnest at caput → **more distal anastomosis = easier, better pregnancy**. Adaptation up to 2 years.
- **Indications**: reversal per fluid criteria; non-vasectomy OA with proven spermatogenesis + no vasal sperm + patent vas/ejaculatory ducts.

ALGORITHM

Vas hemitransected at straight–convoluted junction, fluid sampled, patency confirmed → open tunica vaginalis → site **above** obstruction where tubules dilated (else puncture distal → proximal until sperm) → 3–4 mm tunic window → vas through tunica vaginalis, **fixed with 5-0 Prolene to albuginea (tension relief)** → posterior tunic–vas 8-0 → intussusception sutures → outer tunic–vas adventitia

Technique	For	Against
Longitudinal intussusception (LIVE) — used for all	Two 10-0 longitudinal sutures placed in the **still-distended** tubule; incise between; tubule pulled into vas; bloodless, leakproof	Fluid not checked before sutures set
End-to-side	Bloodless; fluid checked first	Suturing a collapsed tubule
End-to-end	Rapid level finding; bridges gap	Haemostasis hard; inferior epididymal supply lost

- Intussusception works because needles go in before collapse and forward flow pushes the tubule against vasal mucosa.
- **No simultaneous varicocelectomy** — stage after 6 months.
- Late shutdown occurs → **bank sperm intraoperatively and once they appear**. Failure → redo or retrieval + ICSI.

7 · TURED

- P3 Ejaculatory duct obstruction — *congenital (midline cyst, atresia) vs acquired; low volume, acidic, low fructose, palpable vas, normal FSH; TRUS; TURED only after couple declines IVF; sperm return in ~half; cold-knife reobstructs; reflux, epididymitis, retrograde ejaculation; vesiculoscopy*

ALGORITHM

24-Fr resectoscope; finger in rectum → resect the **verumontanum** to open dilated orifices/cyst → spare bladder neck, sphincter, rectum → **blue efflux confirms**; minimal coagulation → catheter overnight, 7 days antibiotics

- Avoid confirmatory high-pressure vesiculography (secondary epididymal obstruction).
- Reflux: document by VCUG or semen creatinine. Epididymitis: low-dose antibiotics until pregnancy; chronic → vasectomy/epididymectomy. Retrograde ejaculation: pseudoephedrine 60 mg 90 min before, or alkalinised urinary sperm.

8 · Sperm retrieval

- P2 Sperm retrieval techniques (sperm extraction in obstructive azoospermia; testicular aspiration and assisted reproduction) — *ICSI essential; PESA/MESA/TESA/TESE/micro-TESE table; MESA by capillary action; micro-TESE ~half of NOA; retrieval by diagnosis (AZFa/b **0% — do not operate**); frozen = fresh except SCI*
- MESA: highest motile numbers, multiple vials. PESA: pregnancy ~half of open. TESE: most consistent open technique. **Micro-TESE = best in NOA**.

KEY POINT
Paradox: in obstruction the **most motile sperm are proximal** (efferent ductules); distal tubules hold senescent fragments and macrophages. Yet fertilisation is best when the longest epididymis is preserved.

ALGORITHM
Micro-TESE: general/regional → 10× avascular plane on anterior albuginea → wide equatorial incision → bivalve → excise larger tubules → in-theatre andrologist examines → map positives → negative: contralateral → close albuginea 5-0, tunica vaginalis 4-0

9 · Cryopreservation

- Freeze at diagnostic biopsy, every reconstruction, and from ejaculate once motile sperm return after reversal. Sample before any dye; epididymal vials allow elective retrieval.

10 · Varicocelectomy

- P2 P3 Surgical management of bilateral varicocele; Varicocele: controversies — *indications; approach table (microsurgical recurrence <1%, hydrocele ~0); micro-Doppler, testis delivery, impulse test; complications*
- Most common infertility operation. Repair halts decline, improves semen, live birth, IVF/ICSI outcomes, testosterone; younger → greater benefit.
- Conventional inguinal: hydrocele 3–15%, recurrence 5–15% (misses scrotal/periarterial veins).
- Subinguinal: less pain but more veins, branched artery. **Open external oblique** in prepubertal children, solitary testis, tight low ring, short cord, inexperience or hernia repair; stay subinguinal after inguinal surgery or obesity.

ALGORITHM
Local + sedation → 2–3 cm incision over external ring → cord on Penrose (exclude ilioinguinal, genital branch) → **micro-Doppler maps arteries before handling** → open spermatic fasciae → 10–25×, 1% papaverine → artery looped → **all veins except vasal ligated** → lymphatics preserved → deliver testis: ligate external spermatic, gubernacular veins → impulse test

- **End point**: only testicular and cremasteric arteries, lymphatics, vas with vessels remain. Pulsation is unreliable; artery (1–1.5 mm, ~two-thirds of flow) often under a large vein.
- Ligate vasal veins only if >2.5 mm, keeping one set — **one intact deferential set guarantees venous return**.
- Aftercare: intercourse 1 week; activity 4 weeks (6 if fascia opened). Recurrence: venography + embolisation.

KEY POINT

Recurrence = a vein missed; hydrocele = a lymphatic tied; atrophy = an artery tied. Microsurgery with testicular delivery addresses all three.

11 · Landmark trials

Trial	Design	Result	Take-home
Madgar (1995)	RCT: varicocelectomy vs no surgery in infertile men	1-year pregnancy 44% vs 10%	Repair of clinical varicocele with abnormal semen improves fertility

71 Physiology of Penile Erection and Pathophysiology of Erectile Dysfunction

1 · Definition and key concepts

- **Erectile dysfunction (ED)**: consistent or recurrent inability to attain and/or maintain an erection sufficient for sexual satisfaction (AUA 2018); replaces "impotence".
- Erection = **sinusoidal (trabecular) smooth muscle relaxation + arterial dilation + venous compression (veno-occlusion)**; a neurovascular event — rigidity comes from *trapping* blood, not inflow alone.
- Milestones MCQ: Leonardo da Vinci (hanged men) – erection is blood; 1990s – NO; 1998 – oral PDE5 inhibitors (PDE5Is).

2 · Functional anatomy

Corpora

- Paired dorsal **corpora cavernosa** + ventral **corpus spongiosum** (around urethra, expands as glans).
- Crura arise separately from puboischial rami; septum **incomplete** in humans → one injection fills both corpora.
- Sinusoids (larger centrally) separated by smooth muscle trabeculae; helicine arteries and terminal cavernous nerves lie close to trabecular muscle.
- Sinusoidal Po_2: flaccid venous **~35 mm Hg**; erect arterial **~90 mm Hg**.

Tunica albuginea

- **Inner circular layer** holds cavernous tissue, gives **intracavernous pillars** (absent 5–7 o'clock). **Outer longitudinal layer** glans→crura, thicker dorsally, absent in ventral groove.
- Ventral ~0.8 mm vs dorsal ~2.2 mm → ventral (5–7 o'clock) weak zone: **penile fracture**, implant crossover/perforation.
- **Veno-occlusion**: emissary veins run obliquely between layers and are compressed as sinusoids stretch the tunica; cavernous arteries escape in a periarterial sheath.
- Spongiosal tunica thin, single layer, no struts, absent in glans → spongiosum/glans are **low-pressure AV shunts**, never rigid.

Arteries

- Internal iliac → **internal pudendal** → common penile → **cavernous** (main corporal supply; **helicine arteries** – tortuous flaccid, straight erect, open into sinusoids without capillaries), **dorsal** (glans; perforators join cavernous distally), **bulbourethral** (bulb, spongiosum).
- **Accessory pudendal arteries** (external iliac, obturator, vesical, femoral) in ~one-third; may be dominant — preserving at radical prostatectomy speeds erectile recovery.

Veins

Sinusoids → **subtunical venous plexus** → **emissary veins** through the tunica.

System	Drains	Destination
Deep dorsal vein (± circumflex, periurethral)	Glans + distal two-thirds of corpora	Periprostatic (Santorini) plexus

System	Drains	Destination
Cavernous and crural veins	Proximal (infrapubic) penis	Internal pudendal veins
Superficial dorsal vein	Skin, subcutaneous tissue	Saphenous veins

KEY POINT
Arterial and venous anatomy vary widely between men — so the same "nerve-sparing" or pelvic operation gives different erectile outcomes.

3 · Haemodynamics: phases of erection and detumescence

- P1 Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED — *sequence of events and phase table (flaccid → latent → tumescence → full → rigid → detumescence)*
- Full erection plateau ICP **~100 mm Hg**; rigid phase suprasystolic. Cavernous (autonomic) nerve drives phases 1–3; pudendal → ischiocavernosus drives phase 4.
- Rigid phase: arterial inflow near zero/reversed. Detumescence: 5 initial (transient ICP rise, muscle contracts against closed veins), 6 slow (veins reopen gradually), 7 fast (outflow restored).
- **Trap**: arterial inflow peaks in the latent phase, not at rigidity — basis of Doppler timing (PSV measured early).
- Spongiosum/glans: no veno-occlusion → pressure one-third to one-half of corporal; extra engorgement in rigid phase from bulbo-/ischiocavernosus and dorsal vein compression under Buck fascia.

4 · Neuroanatomy

Peripheral and spinal

	Sympathetic	Parasympathetic	Somatic
Origin	**T11–L2** (often T10–T12)	**S2–S4** intermediolateral columns	S2–S4; motor from **Onuf nucleus**
Route	Chain → superior hypogastric plexus → hypogastric nerves → pelvic plexus	Pelvic nerves (nervi erigentes) → pelvic plexus	Pudendal nerve; afferent via dorsal nerve of penis
Role	**Detumescence**; emission	**Erection** (cavernous nerves)	Sensation; ischiocavernosus (rigidity), bulbocavernosus (ejaculation)

- **Cavernous nerves** (pelvic plexus, both autonomic types) †run posterolateral to prostate; lateral branch pierces urogenital diaphragm 4–7 mm lateral to sphincter. Pelvic ganglia along bladder–prostate junction and seminal vesicles → vulnerable in radical pelvic surgery.
- Dorsal nerve: somatic sensory with some nNOS fibres.
- **Bulbocavernosus reflex**: glans squeeze → bulbocavernosus contraction; limited value in ED work-up.

Three types of erection

Type	Trigger	Mechanism	Spinal cord injury (SCI)
Psychogenic	Audiovisual, fantasy	Brain modulates spinal centres, chiefly by suppressing sympathetic tone	May persist with sacral lesions; absent with lesions above T9

Type	Trigger	Mechanism	Spinal cord injury (SCI)
Reflexogenic	Genital touch	Sacral reflex arc → cavernous nerves	Preserved in upper cord lesions
Nocturnal	REM sleep	Cholinergic pontine tegmentum on; locus coeruleus (noradrenergic) and raphe (serotonergic) silent	—

Supraspinal (central)

- **MPOA** – partner recognition, integrates hormonal/sensory cues; dopaminergic. **PVN** – pro-erectile **oxytocin** neurons to lumbosacral nuclei. Nucleus paragigantocellularis – **serotonergic brake**.
- Pyriform cortex lesions → hypersexuality.

Central neurotransmitters

- P1 Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED — *promoters (dopamine, oxytocin, NO, noradrenaline, melanocortins, 5-HT2C) vs inhibitors (5-HT1A/2, GABA, opioids, cannabinoids, prolactin)*
- Apomorphine = D1/D2 agonist (PVN → oxytocin → NO); MC4 receptor for melanocortins; 5-HT1A/5-HT2 inhibit erection but facilitate ejaculation.

> **KEY POINT**
> NOS inhibitors block both apomorphine- and oxytocin-induced erections → NO is the final common central pathway. **Trap**: noradrenaline is pro-sexual centrally but the principal anti-erectile transmitter peripherally.

5 · Molecular mechanisms of smooth muscle tone

Contraction (flaccid state)

- Corporal smooth muscle is **contracted at rest**: myogenic + adrenergic tone + endothelial constrictors.
- **Ca^{2+} pathway**: noradrenaline (α1), endothelin-1 (ET_A), angiotensin II (AT1), PGF2α, thromboxane A2 → Gq → PLC → IP_3 → Ca^{2+} release + entry → Ca^{2+}–calmodulin → **MLC kinase** → phosphorylates MLC20.
- **Ca^{2+} sensitisation – RhoA/Rho-kinase** inhibits **MLC phosphatase (MLCP)** → sustained tonic contraction *without* further Ca^{2+} rise; latch state (calponin) keeps basal tone cheaply.
- Rho-kinase ↑ in DM, hypertension, hypercholesterolaemia, hypogonadism, ageing, nerve injury; drives fibrosis (fibroblast → myofibroblast).

Relaxation (erection)

- **NO–cGMP**: NOS converts L-arginine + O_2 → NO + citrulline. **nNOS initiates, eNOS sustains**. NO → soluble guanylyl cyclase (haem-dependent) → cGMP → **PKG-I** → K^+ efflux, ↓ Ca^{2+} entry and release, Rho-kinase inhibited, MLCP activated → relaxation.
- **cAMP** (PGE1 via EP2/EP4, VIP, CGRP → Gs → PKA) is **NO-independent** — why intracavernosal alprostadil works when nerves/endothelium fail. Prostacyclin (PGI2) contracts, PGE1 relaxes.
- **PDE5 (cGMP-specific) is the principal cavernous PDE**; PDE6 retinal (visual side effects); cAMP degraded by PDE2, 3, 4. Papaverine nonspecific. ↑ cGMP inhibits PDE3 → ↑ cAMP.

- K^+ channels (maxi-K, K_{ATP}) hyperpolarise. **Gap junctions (connexin-43)** synchronise relaxation → uniform erection from local input.
- O_2 switch: flaccid low Po_2 suppresses NOS, favours endothelin; erect Po_2 >50–60 mm Hg permits full NOS activity.
- **Adenosine** (A2 → cAMP): excess with ↑ A2B signalling → prolonged erection in adenosine deaminase-deficient and sickle cell mice — the **priapism models**.
- **Sonic hedgehog (SHH)** maintains corporal architecture; ↓ after nerve injury and in DM → atrophy.
- Acetylcholine acts indirectly (endothelial NO, presynaptic adrenergic inhibition); **atropine does not abolish** neural erection MCQ.
- PKG-I knockout mice fail to erect; nNOS/eNOS knockouts often erect (redundancy) → PKG is the bottleneck.

Second messenger	Activators	Kinase	Degraded by	Drug
cGMP	NO, CO, H_2S, natriuretic peptides	PKG-I	PDE5	PDE5Is
cAMP	PGE1, VIP, CGRP, adenosine (A2)	PKA	PDE2, 3, 4	Alprostadil; papaverine (both)

Detumescence

NO release stops + PDE5 hydrolyses cGMP + sympathetic noradrenaline (α1; principal transmitter of flaccidity) + endothelin-1 + Rho-kinase → trabecular contraction → veins reopen (phases 5–7).

> **KEY POINT**
> Corporal tone is a tug-of-war over one phosphorylation site (MLC20): Ca^{2+}–calmodulin–MLC kinase and Rho-kinase keep it phosphorylated; NO–cGMP–PKG removes the phosphate. **Trap**: "MLCP" = the phosphatase.

6 · Classification of ED

Approach	Categories
Neurovascular mechanism	Failure to **initiate** (neurogenic, psychogenic, endocrine) · to **fill** (arterial) · to **store** (veno-occlusive)
Aetiological	Psychogenic · neurogenic · hormonal · arteriogenic · cavernosal (venogenic) · drug-induced · systemic disease · mixed (commonest)

Most organic ED has a psychogenic component.

7 · Risk factors and endothelial dysfunction

- **Age** = strongest demographic predictor, but DM, heart disease, hypertension carry more risk (MMAS).
- **Endothelial dysfunction**: oxidative stress oxidises BH4 → **NOS uncoupling** → superoxide → peroxynitrite; arginase depletes L-arginine; ADMA inhibits NOS → ↓ NO.
- **Artery-size hypothesis**: smaller penile arteries become symptomatic first; ED preceded CAD by a mean of **~2 years** → ED is a **sentinel**; stratify cardiac risk → *Ch 72*.

- Common end-path: ↑ Rho-kinase, ↑ TGF-β → collagen, ↓ smooth muscle:collagen → veno-occlusive failure.

8 · Pathophysiology by type

Psychogenic

- Supraspinal inhibition of spinal centres + excess sympathetic tone (higher serum noradrenaline).
- Performance anxiety, partner conflict, depression, PTSD, problematic pornography use.

Neurogenic

- **Brain**: Parkinson, stroke, tumours, dementia, temporal lobe epilepsy, multiple system atrophy, head injury. **Cord**: SCI (commonest), spina bifida, disc disease, MS.

Complete lesion	Reflexogenic erection	Psychogenic erection
Upper cord (above sacral centre)	Preserved in **~95%**; brief, need continuous stimulation	Lost
Lower cord (sacral)	Preserved in only **~25%**	May occur (thoracolumbar pathway); marginal tumescence

- **Iatrogenic**: radical prostatectomy, cystoprostatectomy, APR, pelvic radiation; most have at least transient decline despite nerve sparing; worse with baseline ED and age >60. Retroperitoneal sympathetic injury → loss of emission.
- Cavernous nerve injury → ↓ nNOS, apoptosis, fibrosis, ↓ SHH → venous leak: rationale for early penile rehabilitation.
- **Pelvic fracture**: cavernous nerve/pudendal artery injury; with posterior urethral injury, early realignment preserves potency better than delayed anastomosis.

Hormonal

- **Testosterone (T)** governs **libido** more than mechanics but maintains NOS, PDE5 and smooth muscle; castration → ↓ inflow, venous leak. Low libido = commonest sexual symptom of low T.
- AUA: **morning, fasting** T in all men with ED; low T = **<300 ng/dL**; treat only with ≥2 low levels + symptoms. Nocturnal erections need ~200 ng/dL.
- Nocturnal erections androgen-dependent; visual-stimulus erections largely independent MCQ. DHT is the key penile androgen.
- **Hyperprolactinaemia** (prolactinoma, antipsychotics) → ↓ GnRH, low T, ↓ libido, ↓ MPOA dopamine. Low prolactin (<3 ng/mL) also linked to ED.
- Low T:E2 ratio (obesity), hyperthyroidism and hypothyroidism all cause ED.

Vasculogenic – arteriogenic

- Atherosclerotic/traumatic occlusion of hypogastric–pudendal–cavernous tree → ↓ perfusion pressure, slower and softer erection.
- Risks: hypertension, hyperlipidaemia, smoking, DM, obesity, inactivity, pelvic irradiation, blunt perineal trauma (focal stenosis in young men → revascularisation candidate).
- Chronic hypoxia → ↑ TGF-β1, ↑ collagen, ↓ smooth muscle → "inflow failure becomes storage failure".
- Hypertension: ARBs and nebivolol protect penile tissue; hydralazine/amlodipine lower BP without protection.

Vasculogenic – veno-occlusive (venous leak)

- **Tunical**: ageing, DM, Peyronie disease, penile fracture, Peyronie surgery.
- **Fibroelastic trabecular change** (ageing, DM, nerve injury, hypercholesterolaemia) — sinusoids cannot expand enough.
- **Insufficient relaxation**: anxiety, adrenergic tone, endothelial dysfunction.
- **Acquired shunts** after priapism shunt surgery (glans–cavernosum, cavernosum–spongiosum); congenital ectopic/crural veins (primary ED).

> **KEY POINT**
> Veno-occlusion is an emergent property of healthy tissue (elastic tunica + fully relaxed trabeculae), so most "venous leak" is really cavernosal smooth muscle disease — why venous ligation surgery fails and is investigational.

- Vasculogenic ED changes: cavernous artery ↑ resistance; smooth muscle ↑ tone, ↓ content, altered K^+ channels/gap junctions; cavernous fibrosis and elastin loss; impaired endothelium-dependent relaxation; altered tunical fibres; ↓ nNOS/eNOS.

Metabolic and systemic disease

- **DM**: earliest, most severe ED — arteriopathy, nitrergic neuropathy, endothelial dysfunction, AGEs, ↑ Rho-kinase/arginase, hypogonadism. **Key**: NO *source* damaged but response to exogenous NO preserved → PDE5Is work until fibrosis.
- **MetS/obesity**: visceral fat is endocrine; weight loss, Mediterranean diet, exercise, bariatric surgery help.
- **Hyperlipidaemia**: oxidised LDL inhibits NO; statins linked to better IIEF.
- **CKD**: up to **80%** of ESRD men have ED; persists in ~half after transplant.
- Others: HIV, cirrhosis, COPD, OSA, BPH/LUTS.
- **Ageing**: ↑ latency, ↓ rigidity, ↓ ejaculatory force, longer refractory period, ↓ nocturnal erections.

Drug-induced

Class	Cause ED	Mechanism / note	Alternative
Antihypertensives	**Thiazides**, **nonselective β-blockers**, clonidine	Thiazide – unknown; β-blocker – prejunctional α2	ACEIs, ARBs, nebivolol, α1-blockers (ejaculatory dysfunction), CCBs
Antidepressants	SSRIs (paroxetine worst), tricyclics	5-HT_2/5-HT_3 spinal inhibition, ↓ dopamine, ↓ NO; post-SSRI dysfunction may persist	Bupropion, mirtazapine, nefazodone, buspirone; add PDE5I
Antipsychotics	Typical > atypical	Dopamine block → hyperprolactinaemia	Newer agents
Antiandrogens	LHRH agonists/antagonists, AR antagonists, cyproterone, **5ARIs**	5ARI: least T effect yet persistent ED after stopping	None
Recreational	Tobacco (dose–response, ≥20 pack-years), heavy alcohol, opioids	Endothelial damage, neuropathy, hypogonadism	Stop/moderate

Class	Cause ED	Mechanism / note	Alternative
Others	Cimetidine, spironolactone, carvedilol	—	—

Trazodone (m-chlorophenylpiperazine) → priapism. Anticonvulsants: valproate ↓ desire; carbamazepine/phenytoin worse than lamotrigine.

Primary (lifelong) ED

- Usually psychogenic; exclude hypoplastic cavernous arteries, congenital venous leak, corporal atresia, megalourethra.
- Micropenis (stretched length >2.5 SD below mean) and curvature may present as "ED".

> **KEY POINT**
> Why ED is a sentinel: nearly every systemic disease reaches the penis through the same three doors — endothelial NO failure, Rho-kinase-driven contraction, and TGF-β-driven fibrosis.

72 Evaluation and Management of Erectile Dysfunction

1 · Definition and significance

- **ED** (AUA): consistent or recurrent inability to attain and/or maintain an erection sufficient for sexual satisfaction; minimum duration **3 months** (except trauma/surgery-induced ED).
- Prevalence ~**20%** of adult men (40–49 y up to 15%; 60–69 y up to 40%). One in four new presenters is <40 y, half severe.
- Opioid use doubles risk; prior COVID-19 ~sixfold.
- **ED predicts CVD** (shared endothelial dysfunction) — comparable to smoking or family history of MI; linked to all-cause mortality. Younger onset → greater CV risk.

Independent association (multivariate)	Risk factors
Strongest	Antidepressant use, then antihypertensive use
Strong	DM (strongest disease); obstructive urinary symptoms
Moderate	Hypertension, prostate enlargement, smoking, BMI ≥30, inactivity
Weak/null after adjustment	CVD, hypercholesterolaemia

- **Drug-induced ED** up to 25% of presentations → *Ch 71*.

2 · Management principles

- **Shared decision making** (AUA 2018): all non-contraindicated options (PDE5Is, VED, intraurethral (IU) alprostadil, intracavernosal injection (ICI), prosthesis) presented **in parallel, not ranked**; patient may choose any first. Older **step-care / Process of Care Model (1999)** ranked oral → local/VED → prosthesis.
- T deficiency (symptoms + total T <300 ng/dL): restoring T likely improves other ED treatments.
- **Partner interview** changes diagnosis/treatment in over half in some series; drives adherence. **Follow-up** reduces high discontinuation.

Cardiac risk assessment – Princeton Consensus

- P1 Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED — *Princeton high (defer sex and ED treatment) / indeterminate (stress test, restratify) / low (treat)*

Every man with ED is at potential CV risk; sex ≈ moderate exertion. Framingham/global risk score is a useful first step.

Refer: failed therapy, young trauma patient, deformity, medicolegal, complex endocrine/psychiatric/vascular problems.

3 · Basic evaluation

History

- Onset, situational vs global, morning/nocturnal and masturbatory erections, libido, ejaculation, curvature, partner factors, prior treatment; is ED secondary to another phase (e.g. premature ejaculation)?

- Medical: DM, CVD, neurological disease, pelvic surgery/radiation, trauma, drugs, depression (bidirectional; Whooley two-question screen).

Psychogenic	Organic
Sudden onset	Gradual onset
Complete, immediate loss	Incremental progression
Situational	Global
Waking/nocturnal erections present	Poor or absent

> **KEY POINT**
> Preserved nocturnal/morning erections prove an intact neurovascular axis — the strongest historical pointer to psychogenic ED.

Questionnaires

- **IIEF**: 15 items, 5 domains (erectile function, orgasm, desire, intercourse satisfaction, overall satisfaction).
- P1 Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED — *IIEF-5/SHIM bands: 5–7 severe … 22–25 no ED*
- SHIM: 5 items scored 1–5, past 6 months †. Questionnaires grade severity and response but do **not** identify cause.

Physical examination

Testes/gynaecomastia/virilisation (hypogonadism); BP, obesity, pulses (vascular); genital sensation, bulbocavernosus reflex (neurogenic); plaque, micropenis, chordee (structural).

Laboratory tests

- **Recommended**: fasting glucose/HbA1c, lipids, serum chemistry, FBC, **morning (7–11 am) total testosterone**.
- Low T → repeat (≈one-third of mildly low values normalise) with LH and prolactin; thyroid tests at discretion; PSA if prostate disease suspected or before T therapy.

> **KEY POINT**
> The basic work-up finds the disease behind the ED — undiagnosed DM, dyslipidaemia, hypogonadism, CV risk — not proof of ED.

4 · Hormonal evaluation

- P4 Male menopause (andropause, late-onset hypogonadism) — *bioavailable T = free + albumin-bound; SHBG rises with age; LH separates primary/secondary; prolactin if LH low; pituitary MRI if T <150 ng/dL*
- Free T 0.5–3%, SHBG-bound ~30%, albumin-bound ~67%.
- SHBG ↓: obesity, nephrotic syndrome, hypothyroidism, glucocorticoids. SHBG ↑: ageing, cirrhosis, hyperthyroidism, HIV, anticonvulsants, oestrogens. In DM use bioavailable T.
- AUA threshold **300 ng/dL (10.4 nmol/L)**; sexual function correlates weakly with T above ~230 ng/dL. Pituitary MRI if T **<150 ng/dL**.
- Hyperprolactinaemia: antipsychotics, tricyclics, opiates, prolactinoma, hypothyroidism, renal failure. Oestradiol if gynaecomastia.

5 · Specialised testing

For young trauma patients, primary ED, surgical candidates, medicolegal cases, failed therapy.

Test	Line	Answers	Indication
Combined intracavernosal injection and stimulation (CIS)	1st	Can vessels produce a rigid erection?	Screening
Penile duplex Doppler	2nd	Arterial vs veno-occlusive; asymmetry	Most reliable, least invasive
DICC (dynamic infusion cavernosometry/cavernosography)	3rd	Venous leak and where?	Trauma, lifelong ED, before venous surgery
Selective internal pudendal arteriography	3rd	Arterial lesion site; inferior epigastric donor	Young trauma patient before revascularisation

CIS test

- Alprostadil 10–20 µg or Bimix/Trimix 0.3 mL, 27–29 G needle, lateral base; compress 5 min; stimulate; rate rigidity.
- Rigid sustained erection = normal haemodynamics. False negatives: anxiety, low dose (redose); false normal in up to a fifth with borderline inflow.
- No detumescence by ~1 h → intracavernosal phenylephrine (diluted 500 µg/mL) every 3–5 min.

Penile duplex Doppler

- Cavernous artery PSV/EDV at the base every 5 min to 20 min after ICI; redose if poor erection.

Parameter	Arterial insufficiency	Veno-occlusive dysfunction
PSV	**<25 cm/s** (25–35 borderline; >35 normal)	Persistently >25 cm/s
EDV	—	**>5 cm/s**
RI = (PSV − EDV)/PSV	—	**<0.75** (normal >0.9)
Other	Acceleration time >122 ms; side-to-side PSV difference >10 cm/s; artery dilation <75%	Rapid detumescence
When assessed	Within 5 min	15–20 min (needs full relaxation)

> **KEY POINT**
> Venous leak can be diagnosed only with adequate inflow and full smooth muscle relaxation — an anxious patient mimics veno-occlusive dysfunction, so redose first. Anatomical variants (dorsal–cavernous communications, early branching) can falsely suggest arteriogenic ED.

DICC and arteriography

- DICC: vasodilator → saline infusion → flow-to-maintain, pressure decay, cavernous artery systolic occlusion pressure (CASOP) → cavernosography localises leaks. Veno-occlusive dysfunction = cannot reach mean systolic pressure or rapid fall after infusion stops.
- Obsolete: penile–brachial index, plethysmography.

Nocturnal penile tumescence and rigidity (NPTR)

- Sleep-lab NPTR with polysomnography is the standard.
- **RigiScan** (home): base and coronal loops record tumescence and **radial** rigidity; normal = 4–5 erections/night, ≥ †60% rigidity lasting ≥10 min. Cannot verify REM; radial ≠ axial.

- Use: obscure aetiology, medicolegal, suspected psychogenic ED. False normal in MS; false abnormal in ageing, depression, anxiety.

Neurological tests (bulbocavernosus reflex latency, dorsal nerve conduction, biothesiometry): limited role. Penile neuropathy in DM can exist without foot neuropathy.

6 · Treatment

ALGORITHM

Treat reversible causes (lifestyle, drugs, hormones, psychosocial) + CV risk stratification → shared decision: PDE5I | VED | IU alprostadil | ICI | (Li-ESWT investigational) → inadequate response: optimise, combine, switch → penile prosthesis → *Ch 75*

Lifestyle and risk-factor modification

- Smoking cessation, exercise, weight loss, Mediterranean diet improve function; control DM, lipids, BP.
- **Medication change**: thiazides/β-blockers → CCBs, ACEIs, ARBs; SSRI → bupropion, mirtazapine, buspirone, drug holiday, or add PDE5I.
- **Psychosexual therapy** (sensate focus, CBT, couples) — best combined with medical therapy.

Hormonal therapy

- T for confirmed deficiency: ~3-month trial; improves libido more than erection; synergistic with PDE5Is.
- Monitor DRE, PSA, haematocrit, LFT, lipids at baseline → 3–6 months → yearly. AEs: erythrocytosis, sleep apnoea, acne, gynaecomastia, LUTS. RCT meta-analysis: no causal CV link.
- Preparations: gels (transfer risk), patches, IM enanthate/cypionate (200–250 mg every 2–3 weeks; supraphysiological peaks, polycythaemia), undecanoate depot (750 mg at 0, 4 weeks, then every 10 weeks), pellets, nasal gel (no transfer), oral undecanoate (raises BP). Avoid 17α-alkylated methyltestosterone (hepatotoxic).
- Hyperprolactinaemia: T not corrective — stop drug; bromocriptine for prolactinoma; surgery for failure/visual compromise.

PDE5 inhibitors (first-line oral)

Block cGMP breakdown → **enhance, not initiate** erection; sexual stimulation obligatory.

	Sildenafil	Vardenafil	Tadalafil	Avanafil
Doses (mg)	25, 50, 100	5, 10, 20	5, 10, 20 (2.5/5 daily)	50, 100, 200
T_{max} (h)	0.8 (~1 h before)	0.7–0.9 (~1 h)	**2** (~2 h)	**0.3–0.5** (~30 min)
Onset (min)	15–60	15–60	15–120	15–60
Half-life (h)	3–5	4–5	**17.5**	3–5
Fatty food	↓ absorption	↓ absorption	**No effect**	↓ absorption
Characteristic AE	Visual (blue tinge; PDE6), flushing	Visual, flushing; **QT prolongation – avoid with class 1A/3 antiarrhythmics**, congenital long QT	**Back pain/myalgia** (PDE11)	Fewest visual effects
Nitrates	**Contraindicated (all)**			

- Common AEs: headache (commonest), dyspepsia, flushing, nasal congestion. Efficacy equivalent (~70% successful intercourse; 40–50% in DM and post-prostatectomy).
- **Nitrates** in any form (incl. amyl nitrite "poppers") → profound hypotension; use >2 weeks earlier not a contraindication. If angina: no nitrates for **24 h** (sildenafil, vardenafil) or **48 h** (tadalafil). Hypotension: Trendelenburg → IV fluids → phenylephrine → intra-aortic balloon pump; no antidote.
- Caution: α-blockers (start low, stabilise first), aortic stenosis/LVOT obstruction, hypovolaemia. CYP3A4 inhibitors (ketoconazole, ritonavir) → lower dose; inducers (rifampicin) → higher.
- Not studied: MI/stroke within 6 months, BP <90/50 or >170/100, retinitis pigmentosa, Child-Pugh C, dialysis.
- NAION: no proven increase; stop and seek care for sudden visual loss.
- **Before labelling failure**: adequate stimulation, timing/food, maximum dose, **up to 9–10 attempts**, correct hypogonadism, glycaemia, lipids; switch agent or daily tadalafil.
- **Penile rehabilitation** post-prostatectomy: nightly sildenafil trial positive, REINVENT and nightly-vs-on-demand trials negative — unproven.

Other oral/central agents

Yohimbine (α2 antagonist) – psychogenic only; sublingual apomorphine (D1/D2, nausea; not US-approved); trazodone ≈ placebo; phentolamine – minimal ED only.

Vacuum erection device

- Negative pressure engorges corpora; base ring retains blood — **remove ring within 30 min**. Erection venous, hinged at base, cool, cyanotic.
- Erection up to 90%, satisfaction 30–70%; suits older men in stable relationships; helps soft glans.
- AEs: pain, numbness, trapped ejaculate, petechiae; rarely skin necrosis. Caution with anticoagulants.

Intraurethral alprostadil (MUSE)

- PGE1 pellet 125–1000 µg placed 3 cm into urethra after voiding; massage, stay upright.
- Responders ~50%; niche (PDE5I failure after nerve injury, with PDE5I, soft glans after prosthesis); constriction band (ACTIS) helps.
- AEs: urethral/penile pain (~one-third), minor bleeding, hypotension; priapism rare. Condom if partner pregnant.

Intracavernosal injection

Acts directly on smooth muscle, **independent of nerves and NO** → works when PDE5Is fail (neurogenic, post-prostatectomy, DM).

Agent	Mechanism	Notes
Alprostadil (PGE1) 5–40 µg	Adenylate cyclase → cAMP	Only FDA-approved ICI; ~70% efficacy; 96% metabolised locally within 60 min; **penile pain** commonest AE; least priapism and fibrosis
Papaverine	Nonspecific PDE inhibitor; Ca^{2+} channel block	Cheap; ↑ liver enzymes, priapism, fibrosis → not monotherapy
Phentolamine	α-antagonist	Adjunct only; hypotension, reflex tachycardia

Agent	Mechanism	Notes
Trimix (alprostadil 10 µg + papaverine 30 mg + phentolamine 1 mg per mL)	Synergistic	≈90%; rescues monotherapy failures; less pain
Bimix	Papaverine + phentolamine, or alprostadil + phentolamine	≈90%

- Office titration from low dose (neurogenic/psychogenic very sensitive); inject lateral shaft (2 or 10 o'clock †) avoiding dorsal bundle and urethra; aim ≤1 h erection.
- **Priapism**: highest with papaverine mixes, neurogenic/young men; seek care if >†4 h → aspiration ± phenylephrine. Long-term: fibrosis/curvature.
- Contraindications: psychological instability, priapism risk (sickle cell), severe coagulopathy, unstable CVD, poor dexterity (partner may inject), MAO inhibitors (phenylephrine reversal → hypertensive crisis).

Topical agents

Alprostadil cream (Vitaros) – modest; MD3000 (Eroxon) – first OTC FDA-cleared glans gel.

Restorative and investigational

Therapy	Proposed mechanism	Evidence/status
Li-ESWT	Angiogenesis; nerve, endothelial, smooth muscle regeneration	RCT meta-analysis: ↑ IIEF vs sham in **mild–moderate vasculogenic** ED — best evidence; still investigational (SMSNA: research only)
Platelet-rich plasma	Growth factors	Conflicting RCTs
Stem cells	Regeneration	Uncontrolled series only
Botulinum toxin ICI	Blocks sympathetic noradrenaline	Short-term benefit in refractory ED

Combination therapy

Monotherapy nonresponse approaches 40%. Pairs: daily tadalafil 5 mg + on-demand sildenafil; PDE5I + T; PDE5I + IU/ICI/VED; ICI + VED; prosthesis + IU (soft glans). PDE5I + ICI off-label — test in office.

Surgery

- **Penile prosthesis**: when other therapy contraindicated, ineffective or unwanted; corporal fibrosis after ischaemic priapism/infection; Peyronie disease *→ Ch 75*.
- P2 Surgical management of erectile dysfunction and penile prosthesis — *arterial revascularisation (age <55, nonsmoker, nondiabetic, focal post-traumatic pudendal stenosis; glans hyperaemia) and investigational venous ligation*

7 · Landmark trials

Trial	Design	Result	Take-home
Prostate Cancer Prevention Trial (placebo arm)	Prospective cohort, men ≥55	ED → 45% higher cardiac events at 5 years	ED is a CV sentinel; stratify risk

Trial	Design	Result	Take-home
REINVENT	RCT: nightly vs on-demand vardenafil vs placebo after nerve-sparing prostatectomy	No difference in spontaneous recovery	Scheduled "rehabilitation" PDE5I unproven

73 Priapism

1 · Definition and classification

- **Priapism**: full or partial erection lasting **≥4 h** beyond or unrelated to sexual stimulation. Emergency: fibrosis, shortening, permanent ED.
- Classified by **oxygenation, not duration**: failed outflow → ischaemia (time-critical); excess arterial inflow → oxygenated, painless, can wait.

Feature	Ischaemic (low-flow)	Stuttering (recurrent ischaemic)	Non-ischaemic (high-flow)
Share	**95%**	Subset of ischaemic	<5%
Lesion	Failed outflow, compartment-syndrome-like	Repeated self-limiting episodes, often nocturnal	Cavernosal artery–sinusoid fistula
Typical cause	Drugs, SCD, haematological	SCD	Blunt perineal/penile trauma, needle injury
Pain / rigidity	Painful (within 6–8 h); fully rigid, tender	Painful; rigid during episode	Painless; partially rigid
Blood gas / Doppler	Hypoxic, hypercarbic, acidotic; absent/low flow	Ischaemic per episode	Arterial; high flow, fistula/blush
Management	Emergency decompression	Treat episode as ischaemic + prophylaxis	Observe → embolise
Fibrosis/ED	Common if untreated	Risk with each episode	Usually absent

> **KEY POINT**
> Trap: stuttering priapism is **ischaemic** — episodes ≥4 h need the same first-line treatment.

2 · Aetiology

- P3 Priapism: Diagnosis and Management — *causes: ICI, α-blockers, trazodone, antipsychotics, cocaine, anticoagulant withdrawal, SCD, leukaemia, metastases, venom; non-ischaemic trauma with delayed onset*

Ischaemic – extra points

- ICI misuse/recreational use: younger men, late presentation, often HIV-positive.
- Other drugs: antihypertensives (hydralazine, propranolol), oral PDE5Is, ADHD agents (methylphenidate, atomoxetine), GnRH, testosterone, hydroxyzine, alcohol, marijuana. Take a full drug history.
- Haematological: thalassaemia, myeloma, asplenia, erythropoietin, dialysis with heparin. Leukaemia: cytotoxics + leukapheresis alongside penile treatment.
- **Recurrence after shunt** may be shunt thrombosis: **aspirin 325 mg + heparin 5000 U subcutaneously** pre-op → aspirin 81 mg daily × 2 weeks.

Sickle cell disease

- HbS polymerises on deoxygenation → haemolysis, low NO → outflow obstruction.
- Lifetime risk **29–42%**; 23% of adult and **63% of paediatric** ischaemic priapism. Onset ~15 y; stuttering commoner than major episodes. Precipitants: arousal, sleep, fever, cold, dehydration. Sickle trait: low risk.

Non-ischaemic

- Also neoplastic erosion — men >55 y think malignancy. Delayed onset (≥1 day): clot over the tear disperses in a nocturnal erection.
- Mimic: transient **post-ischaemic hyperaemia** after treatment — colour Doppler distinguishes.

3 · Molecular pathophysiology

- **Failed detumescence**: low basal eNOS-derived NO → ↓ cGMP feedback → **PDE5 down-regulated**; a neural NO surge then gives undegradable cGMP → unopposed relaxation. Also ↓ Rho-kinase, ↑ adenosine (HIF-1α).
- Prolonged hypoxia → apoptosis, TGF-β fibrosis → ED.
- Paradox: **daily PDE5I restores PDE5 and prevents** recurrent episodes.

4 · Evaluation

- P3 P1 Priapism: Diagnosis and Management; Doppler in urology — *history, examination, corporal blood gas table (ischaemic pO_2 <30, pCO_2 >60, pH <7.25; flaccid 40/50/7.35; arterial >90/<40/7.4), PDUS (PSV <50 vs >50 cm/s), parallel labs*
- Glans and spongiosum soft in both types — involvement suggests other pathology. Vitals before sympathomimetics.
- Gas may be skipped after known recent ICI; check colour first (dark vs bright red). MRI: viability (non-acute).

> **ALGORITHM**
> History + examination + gas → ischaemic → acute ischaemic algorithm | indeterminate → penile/perineal PDUS | non-ischaemic → expectant algorithm

5 · Acute ischaemic priapism (AUA/SMSNA)

- Irreversible damage begins 4–12 h after presentation. Counsel on ED risk first. Cold packs, exercise, ejaculation, oral agents, IV fluids: **adjunct only — never delay first-line**. Oral terbutaline/pseudoephedrine not recommended.
- P3 Priapism: Diagnosis and Management — *ED by duration (SCD series): ≤12 h 0%, 12–24 h 22%, 24–36 h 56%, ≥36 h* ***100%***

> **ALGORITHM**
> <36 h: aspiration ± irrigation + intracavernosal phenylephrine (BP and cardiac monitoring) → distal shunt ± tunnelling → tunnelling if not done → proximal shunt / penoscrotal decompression → prosthesis if refractory
>
> >36 h: counsel high ED risk + analgesia/follow-up, OR early penile prosthesis

Aspiration / irrigation

- Penile block → **19-G needle** at penoscrotal junction, **3 or 9 o'clock**. †One side usually suffices (septum fenestrated). Aspirate with shaft compression, repeat, irrigate saline.
- Aspiration alone works in <1 in 4; cold (10 °C) irrigation 96% vs 60% (RCT) — **avoid chilled saline in SCD**.

Phenylephrine

- Selective α1-agonist (minimal β → least cardiac effect).

- Dilute to **100–500 µg/mL**; **1-mL aliquots every 3–5 min** for up to **1 h** before failure; †maximum ≈1 mg in 1 h (EAU).
- Toxicity: hypertension, headache, arrhythmia, **reflex bradycardia** — monitor, especially CVD or MAO inhibitors.

Shunts

- P3 P2 Priapism: Diagnosis and Management; Management of priapism — *Winter, Ebbehoj, T-shunt, Al-Ghorab, snake tunnelling (7/8 Hegar), Quackels/Sacher (stagger ≥1 cm), Grayhack/Barry, penoscrotal decompression*
- Distal shunts create a cavernosum–spongiosum fistula that closes after detumescence; no head-to-head comparisons. Winter = bedside, least invasive, lowest success; T-shunt uses No. 10 blade lateral to meatus, bilateral if needed; Al-Ghorab = transverse incision 1 cm distal to dorsal corona; snake rescues prolonged/failed cases.
- Proximal: Quackels unilateral, Sacher bilateral (1-cm ellipses, 5-0 PDS); Grayhack saphenous vein mobilised ~10 cm; Morey penoscrotal decompression (paediatric Yankauer) is glans-sparing and permits concurrent prosthesis.
- AUA: proximal shunts only after failed distal shunt with tunnelling. Complications: oedema, haematoma, urethral injury/fistula, necrosis, infection, PE.

Early penile prosthesis

- For priapism **>36 h** or refractory to shunt ± tunnelling; oedematous corpora dilate easily, delayed surgery meets dense fibrosis.
- Early vs delayed: shortening 3% vs 40%, fewer revisions/infections, satisfaction 96% vs 60%; rigidity similar.
- Prior distal shunts raise distal perforation risk — if recovery unlikely, weigh shunting against a future implant → *Ch 75*.

6 · Non-ischaemic priapism

- **Not an emergency**: ~60% resolve with observation, compression, ice.

> **ALGORITHM**
> Observe, reassess at 4 weeks → persists → PDUS (fistula site, size) → counsel → selective internal pudendal/cavernosal artery embolisation → recurrence → repeat embolisation with non-resorbable agent → surgical ligation only after repeated failure

Agent	Type	Feature
Autologous clot	Temporary	Children; early recurrence
Gelatin sponge	Temporary (2–6 weeks)	Reabsorbed
Microcoils	Permanent	Theoretical ED risk
Polyvinyl alcohol particles	Permanent	**Preferred for repeat** (AUA)
N-butyl cyanoacrylate	Permanent	Second line for recurrence

- Embolisation success ≥75%, recurrence 30–40%. Open ligation: ~50% ED, gangrene, abscess.

7 · Stuttering priapism — prevention

- Episode ≥4 h = acute ischaemic priapism. Short episodes: fluids, walking, analgesia. Home self-injected phenylephrine for breakthrough.
- No prophylactic agent has robust evidence.

Agent	Dose	Note
PDE5I	Sildenafil 25–50 mg daily; tadalafil 5–10 mg	Most promising; †start when flaccid; only RCT missed primary endpoint
Pseudoephedrine	60–120 mg at night	Off-label first-line α-agonist
Etilefrine	50–100 mg daily	α1-agonist; Cochrane: insufficient evidence
GnRH agonists, antiandrogens (bicalutamide, flutamide, cyproterone)	Depot / oral	Suppress libido/erections; †avoid before puberty completes
Finasteride	5 → 3 → 1 mg taper	One SCD series
Digoxin	0.25–0.5 mg	Na^+/K^+-ATPase inhibition
Baclofen	Oral 40 mg nocte; intrathecal	Neurogenic/reflex erections

SCD specifics

- Same first-line aspiration/phenylephrine; IV hydration and analgesia **concurrently, never instead**. †Exchange transfusion is not primary treatment (AUA) — involve haematology. Avoid chilled irrigation.

8 · Landmark trials

Trial	Design	Result	Take-home
Ateyah 2005	RCT, ICI-induced priapism, irrigation temperature	10 °C saline 96% vs 60% at 37 °C	Cold irrigation (not in SCD)
Burnett 2014	RCT, SCD, sildenafil vs placebo	Primary endpoint missed; open-label benefit	PDE5I prophylaxis plausible, unproven
Zacharakis 2014	Early vs delayed prosthesis	Shortening 3% vs 40%; satisfaction 96% vs 60%	Implant early when ED is inevitable

9 · MCQ one-liners

MCQ ONE-LINERS

1. First case report of priapism: 1824.

74 Disorders of Male Orgasm and Ejaculation

1 · Spectrum

- Ejaculatory dysfunction (EjD) is the **most common male sexual disorder**: premature (PE) → delayed (DE) → anejaculation; plus retrograde (RE), painful ejaculation, anhedonia, postorgasmic illness syndrome (POIS).
- Orgasm is a separate **cortical** event, so ejaculation and orgasm can dissociate.

2 · Physiology of ejaculation

- Lumbar interneurons at **L3–L5** = **spinal ejaculation generator**, coordinating thoracolumbar and lumbosacral nuclei under genital and supraspinal input.
- Brain: bed nucleus of stria terminalis, medial amygdala, posterodorsal preoptic nucleus, parvicellular subparafascicular thalamus, MPOA, PVN.

Phase	Events	Mediation
Emission	Vas, seminal vesicles, prostate → semen into posterior urethra	Autonomic, **T10–L2**
Ejection	Bulbospongiosus, ischiocavernosus, levator ani rhythmic contraction; **bladder neck closes**; external sphincter relaxes	Somatic **S2–S4 (pudendal)** + sympathetic reflex
Orgasm	Cerebral processing of pudendal input	Cortical

- **Dopamine (D2) promotes**, **serotonin (5-HT) inhibits**; also noradrenaline, ACh, oxytocin, GABA, NO.
- 5-HT2C stimulation delays, 5-HT1A shortens → lifelong PE: **5-HT2C hyposensitivity and/or 5-HT1A hypersensitivity**.

> **KEY POINT**
> Serotonin brakes, dopamine accelerates: PE drugs raise synaptic 5-HT; DE drugs are dopaminergic, noradrenergic or anti-serotonergic.

3 · Premature ejaculation — definition and classification

- P1 P3 Premature Ejaculation — *ISSM 2014 three criteria; lifelong/acquired/variable/subjective; AUA 2 min; aetiology, evaluation (IELT, PEDT, IIEF-5), drug table*

	Lifelong	Acquired	Variable	Subjective
Onset	First encounters, every partner	After normal latency	—	—
Latency	≤ ~1 min (mostly 30–60 s)	Markedly reduced, often ≤3 min	Occasionally short	Normal or long
Basis	Neurobiological, genetic	Medical/psychological	Normal variation	Psychological/cultural
Disorder?	Yes	Yes	No	No

- Waldinger subtypes: variable/subjective dominate the community, lifelong/acquired the clinic.
- ISSM: within **about 1 min** (lifelong) or bothersome reduction often to **≤ about 3 min** (acquired). First **evidence-based** definition; vaginal intercourse only; "about 1 min" allows ~90 s. **Latency alone never suffices**.

	ISSM	DSM-5 (2013)	AUA/SMSNA (2020)
Lifelong latency	≤ ~1 min	≈1 min	≈**2 min** since sexual debut
Acquired	Bothersome fall, often ≤3 min	Specifier only	Marked fall (2–3 min or ≥50%)
Duration/frequency	Always/nearly always	≥6 months, 75–100% of occasions	—
Severity	—	Mild 30–60 s; moderate 15–30 s; severe <15 s	—
Basis	Evidence	Same data	Expert opinion

- AUA 2-min line trades specificity for sensitivity. MCQ ICD-10: only older definition with a number (≤15 s); Masters and Johnson: partner satisfied in <50% of episodes.
- Community stopwatch IELT: positively skewed, median **5.4 min**; ≤1 min statistically abnormal.

4 · Premature ejaculation — aetiology and evaluation

- Lifelong: younger, fewer comorbidities. Acquired: older, metabolic syndrome, DM.
- **PE with ED** (up to half of ED men): "rushing" before a partial erection is lost. Pitfall: PE men mistake post-ejaculatory detumescence for ED → false-positive SHIM.
- **CP/CPPS**: PE is the main sexual symptom; antibiotics for proven bacterial prostatitis lengthen IELT.
- **Hyperthyroidism**: PE tracks suppressed TSH, improves with euthyroidism — routine TSH not indicated. Circumcision: no effect.
- Normal IIEF erectile domain ≥26; IIEF-5 ≥22. Examination desirable in lifelong, **mandatory in acquired**; DRE for prostatitis.

5 · Premature ejaculation — treatment

> **ALGORITHM**
> PE confirmed → secondary to ED/other dysfunction? **Yes** → treat primary cause → **No**: lifelong → SSRI ± behavioural therapy (patient preference); acquired → treat cause ± behavioural/SSRI, attempt withdrawal after 6–8 weeks; variable/subjective → reassurance, education, psychotherapy

- Psychosexual education for all; CBT limited first-line in lifelong but important adjunct in acquired PE.

Drug	Dose / timing	Notes
Dapoxetine	30–60 mg, 1–3 h before	**Only licensed oral PE drug** (>50 countries, not FDA); short half-life; works from first dose; no PDE5I interaction; high discontinuation
Paroxetine (daily)	10–40 mg	**Strongest delay**; off-label
Sertraline / fluoxetine / citalopram	50–200 / 20–40 / 20–40 mg daily	Off-label
Clomipramine (TCA)	12.5–50 mg daily or 3–4 h before	Off-label

Drug	Dose / timing	Notes
Topical lidocaine/prilocaine	20–30 min before (spray ≥5 min)	Condom advised; level 1a evidence
PDE5 inhibitors	30–50 min before	Best with comorbid ED; add-on to SSRI
Tramadol	25–50 mg, 3–4 h before	Low evidence; addiction risk

- **Daily SSRIs**: onset 5–10 days, full effect **2–3 weeks**. Early side effects: fatigue, yawning, nausea, diarrhoea, sweating. Cautions: bipolar (hypomania), bleeding with aspirin/NSAIDs, under-18s (suicidality), sperm DNA fragmentation. **Never stop abruptly**. On-demand off-label SSRIs delay less.
- Topicals: penile hypoaesthesia, partner numbness without condom; metered spray (Fortacin) does not penetrate keratinised skin → minimal transfer.
- PDE5I > placebo and PDE5I + SSRI > SSRI (meta-analyses) → allowed with or without ED.

On-demand dapoxetine	Daily off-label SSRI
Licensed; fewer side effects; little withdrawal; infrequent intercourse	Greater delay; full effect 2–3 weeks; must taper; established relationships

> **KEY POINT**
> The real limiter in PE is adherence, not efficacy — most patients stop SSRIs and topicals (cost, lost spontaneity, "antidepressant" stigma).

6 · Delayed ejaculation, anejaculation and anorgasmia

- DE (DSM-5): marked delay/absence despite adequate stimulation, ~75–100% of occasions, with distress. **No validated cut-off**; most men ejaculate at 4–10 min, so >25–30 min with distress suggests DE.
- **Anorgasmia**: absent perceived orgasm. DE ~25% lifelong, ~75% acquired; rises with age. Hallmark: normal erections, **low subjective arousal**.

Category	Causes
Afferent	Ageing (penile afferents, Pacinian corpuscles)
Psychogenic	Inhibited ejaculation; performance anxiety; religious orthodoxy; **idiosyncratic vigorous masturbation**
Endocrine	Hypogonadism, **hypothyroidism** (DE; hyperthyroidism → PE), hyperprolactinaemia
Neurogenic disease	SCI > diabetic autonomic neuropathy > MS
Neurogenic surgery	Radical prostatectomy, rectal surgery, sympathectomy, aortic aneurysm repair, RPLND
Anatomical (→ RE)	TURP, bladder neck incision
Congenital / infective	Müllerian cyst, Wolffian anomaly, prune belly; urethritis, TB, schistosomiasis
Drugs	**SSRIs** (~7-fold DE risk), TCAs, antipsychotics, α-methyldopa, thiazides, alcohol

- Antipsychotics: DE (dopamine block, prolactin) and RE (α-block).

- **Superior hypogastric plexus** injury at RPLND → dry ejaculation; **nerve-sparing RPLND** preserves it in most. TME with autonomic nerve preservation does likewise.
- After radical prostatectomy: no ejaculation, orgasm preserved; climacturia. After radiotherapy: ↓ volume, weaker orgasm.
- Evaluation: why he stops, masturbatory ejaculation (coital-specific vs global), "ejaculated but no semen?" (RE/obstruction). Examine vasa, DRE, bulbocavernosus reflex; morning T, HbA1c, lipids.

ALGORITHM
Orgasm? **Never/sometimes** → inhibited orgasm (psychosexual therapy) or ageing (reassure, change technique) → **Always** → ejaculate? No → **sperm in post-orgasm urine?** Yes = RE; No = aspermia → failure of emission (neurogenic, drug, metabolic) or ejaculatory duct obstruction (EDO)

	Bilateral EDO	RE	Failure of emission
Antegrade volume	Very low	Low/absent	Absent
pH, fructose	Low (no seminal vesicle fluid, ~80% of volume)	—	—
Post-orgasm urine	No sperm	**Sperm + fructose**	No sperm

Treatment

- Aetiology-specific; counsel about anejaculation/sperm retrieval before pelvic surgery. Psychosexual therapist once organic/drug causes excluded (AUA). Normalise T; change offending drug; more stimulation, change masturbatory style.
- Drugs all off-label, ~40% success; poor in psychogenic/neuropathic DE (usually irreversible):

Drug	Action	Dose
Cabergoline	Central D2 agonist	0.5–2 mg every 3 days; **annual echocardiography** (valvular fibrosis)
Bupropion	Dopaminergic	150 mg once/twice daily
Pseudoephedrine	α1 agonist	120 mg 1–2 h before
Reboxetine	Noradrenaline reuptake inhibitor	4–8 mg daily
Cyproheptadine	5-HT antagonist	4–12 mg 3–4 h before (SSRI-induced); sedation
Amantadine, buspirone	—	Anecdotal
Intranasal oxytocin	Oxytocinergic	Both RCTs negative

7 · Retrograde ejaculation

- P3 Retrograde ejaculation — *bladder neck failure; TURP 42–100%, TUIP 5–45%, diabetic neuropathy; post-ejaculate urine; pseudoephedrine/imipramine doses; urinary sperm retrieval → IUI/IVF/ICSI*
- Diagnosis: **>5–10 sperm/high-power field** in post-masturbation urine. TUIP less if one incision not extended to verumontanum; Aquablation superior to TURP for ejaculation in RCTs.

- Antimuscarinic + sympathomimetic better than either.

8 · Sperm retrieval in SCI

- Reflexogenic erection needs the sacral arc (preserved in UMN lesions); ejaculation needs thoracolumbar outflow — rare with complete UMN (<5%). RE common.

> **ALGORITHM**
> **Penile vibratory stimulation (PVS)** — first line, success up to ~70% (needs intact T12–S1-level reflex) → **electroejaculation (EEJ)** — rectal probe stimulating hypogastric sympathetic efferents, effective and safe → prostate massage → surgical retrieval (TESE/TESA/MESA/PESA; testicular sperm extraction succeeds in most)

- Prepare bladder (empty, instil sperm-wash medium) for RE.
- **Autonomic dysreflexia** (lesions above T6) → pretreat with **nifedipine**, monitor BP.
- Low motility initially, improves with repeated ejaculation; fresh testicular sperm may beat frozen.

9 · Other disorders

Disorder	Key points
Painful ejaculation	Prostatitis/CPPS, urethritis, BPH, seminal vesicle calculi, EDO, α-blockers (least with alfuzosin); often none. Surgery (TURP, TURED, pudendal neurolysis) weakly supported
Haematospermia	Usually benign (see pointer below). Seminal vesiculoscopy if refractory *→ Ch 70*; †evaluate >40 y or persistent
Climacturia	Leak at orgasm after radical prostatectomy. Void first → condom → pelvic floor training → constriction ring → male sling
Ejaculatory anhedonia	Orgasm without pleasure; depression, hypogonadism, hyperprolactinaemia, SSRIs
Post-SSRI sexual dysfunction	Genital anaesthesia, low desire, ED, anhedonia persisting months–years
POIS	Myalgia, fatigue, flu-like state within 30 min of orgasm
Orgasmic headache	Dull bilateral, rising with excitement, sudden intense at orgasm

- P1 Haematospermia — *causes, who needs work-up (≥40 y, persistent, haematuria), PSA/TRUS/MRI, treatment*

75 Surgery for Erectile Dysfunction

1 · Place of surgery and devices

- P2 P3 Surgical management of erectile dysfunction and penile prosthesis; Types of penile prosthesis — *last-line but most effective (satisfaction 80–90%; repair 15% at 5 y, 30% at 10 y); Lue goal-directed approach; malleable vs 2-piece vs 3-piece; device names, expansion, coatings; consent; pre-op preparation; revascularisation*
- Extensive aetiological testing does not change outcome.
- Rods sized 0.5–1 cm shorter than corporal length; fixed girths; mechanical failure exceedingly rare. Two-piece never popular (Ambicor discontinued).
- **Material**: Coloplast polyurethane (Bioflex) — cautery **<35 W**; Boston/Rigicon silicone–fabric–silicone tolerates cautery. Narrow cylinders for small/scarred corpora; length-expanding cylinders correct curvature poorly.

2 · Selection and counselling

- **Consent = expectation management**: rigidity, not the former erection. Commonest disappointment: **shorter erection** (pseudocapsule; worse in Peyronie). Document — patients may be litigious.
- **Phimosis** → circumcise and implant ~6 weeks later; elective circumcision with redundant foreskin → simultaneous (implant first, then circumcise).
- Glucose raised on the day → IV insulin. Nasal MRSA carriers: chlorhexidine bathing + mupirocin 2% × 5 days.

3 · Surgical approaches

Approach	Advantages	Cautions
Penoscrotal	Majority choice; secure subdartos pump; good crural exposure	Urethra at risk — Foley first; reservoir via inguinal canal
Infrapubic (Scott)	Reservoir under vision	Dorsal neurovascular bundle — stay sutures at 10 and 2 o'clock; patient pulls pump down daily
Subcoronal	Rods; allows straightening	Circumcise simultaneously (else lymphoedema)
Ventral penile	Rods without bending; foreskin kept	—

4 · Operative technique (three-piece)

ALGORITHM

Catheter → expose tunica → two 2-0 PDS stay sutures per corpus → longitudinal corporotomy → dilate proximal then distal (Metzenbaum, Hegar/Brooks 7→13 mm) → goalpost test → measure (Furlow) → cylinders → test inflation 1 → close corporotomies → test inflation 2 → reservoir → pump → connect → test inflation 3 → irrigate, close

- Dilators point **dorsolaterally**, away from urethra and septum; gently to ischial tuberosity. **Goalpost test** (equal height, symmetry) detects crossover.
- Cylinder equal to or 1 cm shorter than measured — oversizing → buckling, S-deformity; rear tip extenders adjust. Minimal dilation preserves length in primary implants.

- Pump: dependent subdartos pouch, anterolateral to testis. Malleable girth: two parallel Hegars slightly separated = ideal.

Reservoir placement

- **Retzius**: via linea alba or through the inguinal canal piercing transversalis fascia above the pubic tubercle; bladder empty; iliac vessels ~1 inch lateral.
- Hostile pelvis (radiation, cystectomy, robotic prostatectomy, transplant) → **ectopic**, safe with the **lockout valve**. **Submuscular suprafascial** commonest — fill nearly full (underfilling → migration); exclude intraperitoneal. Others: lateral retroperitoneal, epigastric, subcutaneous.
- Large reservoirs may compress the iliac vein → leg oedema.

5 · Intra-operative problems

- **Fibrotic corpora** (post-priapism, post-explant; pelvic radiation does *not* add difficulty): midshaft corporotomy, cavernotomes (Rossello, Uramix), Otis urethrotome, scar excision; **narrow cylinders**, always two; last resort graft (biologic oversized; synthetic sized to defect).
- **Crossover**: lopsided, unequal measurements → leave large dilator in common cavity, redilate other side laterally; Foley midline between cylinders.
- **Crural perforation** (dilator deeper, lengths differ >1 cm) → **suture sling** (2-0 Prolene through rear tip over corporotomy). Windsock abandoned.
- **Urethral perforation**: at corporotomy → close, proceed. Distal (fossa navicularis; irrigant at meatus) → **abandon that cylinder**, place contralateral, no catheter, replace at 5–6 months.
- **Hypermobile glans** (ventral = **SST deformity**, drooping Concorde nose): glans fixation sutures (3-0 non-absorbable) via hemicircumcising incision; distinguish from short cylinders.
- **Residual curvature**: high-rigidity cylinders (AMS 700 CX, Titan, Infla 10 X) + **modelling** (clamp tubing, bend 90° away 90 s); then plication/incision ± graft; <20° acceptable → *Ch 76*.

> **KEY POINT**
> No catheter after distal urethral perforation: the defect is not closed and the urinary stream draws fluid into the urethra (Bernoulli), so urine does not enter it. Wait 5–6 months for scar to obliterate the false tract. Most intra-operative errors are **prevented by instrument direction** and **detected by symmetry** (goalpost test, midline Foley).

> **KEY POINT**
> After robotic prostatectomy the peritoneum is not replaced; bowel fixes near the internal ring, so a blind inguinal Retzius reservoir can compress bowel — choose an ectopic site or place under vision.

6 · Complications

Infection (1–3%)

- Risks: acute hyperglycaemia, revision (disturbed biofilm), skin/urinary sources, MRSA, scarred corpora.
- AUA prophylaxis: IV vancomycin or 3rd-generation cephalosporin + gentamicin 1 h pre-incision, ≤24 h; extended oral courses unproven. High risk: add antifungal ± piperacillin–tazobactam.
- Coatings (InhibiZone rifampin–minocycline; hydrophilic Titan/Infla dipped) halve infection; **no-touch** → <0.5% (organisms enter via wound).

- Diagnosis: fever, erythema suggestive; pus, exposed device, progressive pain, **pump fixed to scrotal wall** definitive. **Remove ALL components**.

Option	Technique	Caveats
Explant, delayed replacement	Antibiotic irrigation drains × 3 days; reimplant 2–6 months	Fibrosis, loss of several cm; VED meanwhile
Mulcahy salvage	Remove all → antiseptic washes → new drapes → immediate new device	~84%; fails with onset <2 months, aggressive organisms, purulence with cellulitis
Delayed salvage	IV vancomycin + gentamicin, salvage at 3 days	Aggressive infection
Malleable salvage	Rods instead	Quick; convert later

- Washes: dilute povidone-iodine (0.35–3.5%) 3 min, then saline/antibiotic; **drop hydrogen peroxide**.
- **Revision washout**: uninfected revision ~10% infection → ~3% with antiseptic washout of all cavities.

Mechanical failure

- **Commonest complication**: leak, pump malfunction, autoinflation, kinking, cylinder aneurysm.
- Device >3 years → replace **all**; retained parts checked by **ohmmeter test**. Rod loss of rigidity → core fracture → replace.

Extrusion and erosion

Feature	Extrusion	Erosion
Position	Tip through tunica, still under skin/glans	Through skin or into urethra
Cause	Forceful dilation, oversizing, aggressive use	Infection, crossover, catheterisation; rods in insensate (SCI) men
Management	**Distal corporoplasty** (new dorsomedial plane behind pseudocapsule)	Remove all, washout, contralateral spacer, replace at 5–6 months

- **Reservoir erosion**: bladder → exchange reservoir, repair; bowel → remove whole device.
- **Penile necrosis** (DM, vascular disease, smoking, prior implant, radiation): avoid circumferential incisions; wrap with glans visible; remove cylinders promptly.
- Early basal bulge = corporotomy disruption (repair); late = cylinder aneurysm (replace).

7 · Special situations

- **Post-priapism** >36 h → **immediate implant** preserves length, relieves pain. Biopsy necrosis; close distal shunt defects; avoid distally expanding cylinders (LGX, Infla 10 AX) → *Ch 73*.
- **Peyronie**: more length loss; non-expanding high-rigidity cylinders + modelling.
- Transplant and HIV-positive men (adequate CD4): no excess infection. Lax suspensory ligament → tunica to symphysis periosteum. Neophallus: single cylinder anchored to pubis, high erosion.

8 · Post-operative care

- Day surgery; cylinders **partially inflated** (large pseudocapsule); mummy wrap; drain <24 h; penis upward 4–6 weeks. Analgesia: dorsal nerve block (liposomal bupivacaine), short opioid, NSAIDs.
- Activation at **4–6 weeks**, partner present; daily cycling. Partners slightly less satisfied; dissatisfaction = length, rigidity.

9 · Penile revascularisation †

- †Young (<55) post-traumatic focal pudendal occlusion only: inferior epigastric → dorsal artery (or deep dorsal vein) — see P2-039 pointer above. Generalised atherosclerosis and venous-leak surgery not recommended.

76 Diagnosis and Management of Peyronie Disease

1 · Definition

- **Peyronie disease (PD)** (AUA 2015): acquired penile abnormality with **fibrosis of the tunica albuginea** ± pain, deformity, erectile dysfunction (ED), distress.

> **KEY POINT**
> Myths corrected: PD is **not rare** (3–20%; higher in diabetes, ED); **spontaneous resolution is rare** — up to half worsen untreated ("wait 6–12 months" is a myth); teens to 70s, all races; flaccid trauma can trigger it; **calcification is not a marker of chronic disease**.

2 · Anatomy and pathophysiology

- Tunica: type I collagen + elastin; **inner circular, outer longitudinal layers**, 1.5–3 mm; septum anchored to inner layer.
- Longitudinal layer thinnest at 3 and 9 o'clock, **absent 5–7 o'clock**; **60–70% of plaques are dorsal** with septal involvement.
- **Microtrauma**: coital bending → **delamination at septal insertions**.
- **TGF-β1** central (+ PDGF, FGF, ROS, iNOS, PAI) → disorganised collagen, ↓elastin → **curvature toward the plaque**.
- Needs a **susceptible host** (penile fracture rarely causes PD); associated **Dupuytren, Ledderhose**. Only 16–40% recall trauma.

3 · Clinical features and phases

- Pain, deformity (curvature, indentation, **hourglass**, hinge, shortening), plaque, ED. Hourglass/hinge → **unstable penis**; distal softness possible.
- Kelami: ≤30° mild, 31–60° moderate, >60° severe. Up to 60° (esp. dorsal) often manageable; ventral/lateral harder.

Feature	Active (acute)	Stable (chronic)
Pain	Present (flaccid, erection, intercourse)	Usually resolved
Deformity	Progressive	No progression
Aim	Prevent progression; NSAIDs for pain; traction	Correct deformity (injection, surgery)
Surgery	Not appropriate	When **≥12 months from onset and stable ≥6 months**

- Differential: congenital curvature, chordee without hypospadias, epispadias; post-traumatic fibrosis; tumours (epithelioid sarcoma mimics PD).

4 · Evaluation

- **Key question**: "If straight with current rigidity, adequate for penetration?" — **No** → prosthesis + straightening; predicts post-op erectile function.
- **PDQ**: 15 items, 3 domains (psychosexual, pain, bother); no total score.
- **Examination**: plaque on stretch (size unreliable). **Stretched penile length (SPL)** pubis to corona dorsally at 90° — essential baseline (patients overestimate loss).

- **Calcification** (US: hyperechoic, shadowing): extensive → poor intralesional response, more surgery/prosthesis.

Levine calcification grade	Size	Surgery likelihood
1	<0.3 cm	Not increased
2	0.3–1.5 cm	Not increased
3	>1.5 cm, or multiple ≥1 cm	Increased

- **ICI ± penile duplex Doppler (PDDUS)**: AUA — required **before any invasive treatment**; goniometer curvature, indentation, hinge, calcification, PSV/EDV/RI.
- Ventral curvature most associated with veno-occlusive dysfunction.
- Morning testosterone only if ED; biothesiometry if sensory change.

5 · Non-surgical management

- P1 Management of Peyronie's disease — *phases, evaluation, non-surgical options table, surgical algorithm*
- No pain, no penetration difficulty → **reassurance**.
- **Oral**: AUA "should not offer" any except **NSAIDs for pain**. Vitamin E, tamoxifen, colchicine, carnitine — no benefit; Potaba shrinks plaque without straightening; pentoxifylline uncertain. PDE5Is treat ED, not PD.
- **ESWT**: pain/function only, not for curvature. **Radiotherapy**: no benefit, real harm.

Collagenase *Clostridium histolyticum* (CCH, Xiaflex)

- Only FDA-approved PD drug (2013). Degrades collagen **types I and III** (despite ↑TIMPs), sparing elastin, vessels, nerves.
- **Protocol (IMPRESS)**: 2 injections of **0.58 mg** 24–72 h apart → **modelling** 24–72 h after the second; up to **4 cycles, 6 weeks apart** (max 8 injections). Modelling essential.
- **AUA criteria**: stable disease, curvature **>30° and <90°**, intact erections (± medication). IMPRESS excluded ventral plaques, extensive calcification, disease <12 months.
- Poor response: calcification, curve ≥60°, severe indentation, hinge. Better: IIEF-5 ≥17.
- Adverse: bruising, swelling, pain; serious — **corporal rupture** (0–4.9%), haematoma → no intercourse until after modelling period.
- Ventral use not routine (urethra shares the collagens).

Other intralesional agents

Agent	Mechanism	Evidence	AUA
Verapamil	CCB: ↓fibroblasts and ECM, ↑collagenase	RCTs conflicting; stabilises; inferior to CCH; cheap	May offer
Interferon α-2b	↓fibroblasts and collagen, ↑collagenase	RCT vs saline: curvature ↓27% vs 9%; flu-like (pre-treat NSAIDs)	May administer
Nicardipine	CCB	↓pain, plaque; no curvature benefit	—

- Verapamil: 10 mg diluted (10–20 mL); poor candidates — calcification, curve >90°, ventral.

Mechanical therapy

- **Penile traction (PTT)**: mechanotransduction. Acute phase 6–9 h/day (off 30 min every 2 h) → less curvature/pain, fewer surgeries, grafting downgraded to plication; gains length; needs

≥3 h/day.

- **RestoreX** (counter-bending, 30–90 min/day): RCT — better length, curvature, IIEF-5.
- Vacuum: modest. AUA: no recommendation for traction, vacuum or combinations.

6 · Surgery

Indications

- **≥12 months from onset, deformity stable ≥6 months**.
- Deformity/inadequate rigidity preventing intercourse; failed conservative therapy.

Consent

- Goal **functionally straight** (≤20°; EAU ≤15°). Residual/recurrent curvature up to 16%.
- **Length loss** with every procedure (plication > grafting); 70–80% already lost length — document SPL.
- **New ED** (grafting > plication), ~30% if pre-op rigidity suboptimal; pre-op erection quality strongest predictor. Sensory loss ~20%, usually transient.

Choice of procedure — algorithm

> **ALGORITHM**
> Erection adequate (± PDE5I)? **No** → inflatable prosthesis + modelling → persistent curve → incision ± graft (graft if defect >2 cm)
> **Yes** → PDDUS → simple curve <60–70°, no hourglass/hinge, acceptable length loss → **plication** (length loss unacceptable → grafting)
> **Yes** → complex/multiplanar >60–70°, severe indentation/hourglass, hinge, short penis (<9 cm stretched) → **plaque incision/partial excision + graft** if ED risk accepted (else prosthesis)

Plication (shortening)

- Shortens the **convex side opposite the plaque**; simple, quick, least new ED. Doesn't fix hourglass/hinge/shortening. Most common PD operation (~half).
- Nesbit failure predictors: poor erections, shortening >2 cm, curvature >30°.

Technique	Method	Point
Nesbit (1965)	Transverse elliptical full-thickness excision opposite curve, transverse closure	Original; ventral curve → lift Buck fascia, dorsal wedges
Yachia	Vertical 0.5–1.5 cm incision closed transversely (Heineke–Mikulicz)	No tunica removed; long incisions narrow shaft
Essed–Schroeder	Non-absorbable figure-of-eight, buried knots	First incisionless plication
16-dot	Permanent extended Lembert, 4 dots per plication	Popular; no incision
TAP (Levine)	Paired partial-thickness transverse incisions (outer layer only); buried permanent + 3-0 PDS	Cavernosal tissue undisturbed

Plaque incision/partial excision + graft (lengthening)

- Complex deformity with **strong erections**, no veno-occlusive dysfunction; restores calibre, regains 0.5–3 cm.
- ED predictors: age >55, veno-occlusive dysfunction (RI <0.80), large graft, ventral curve, curve >60°. Total excision abandoned (ED).

- **Key steps**: degloving; elevate Buck fascia + NVB; H/double-Y incision or partial excision **crossing the septum**, dart corners; graft (pericardium ≤10% oversize; SIS 25%), running 4-0 PDS; plicate residual curve.
- Grafts: pericardium, SIS commonest; synthetics not recommended; outcome depends on selection/technique, not graft.
- Rehab: massage/stretch from 2 weeks; nightly PDE5I; traction ≥3 h/day from 3–4 weeks for 3 months.

Prosthesis with straightening

- PD + PDE5I-refractory ED: one operation. **Inflatable** preferred.
- **Manual modelling** (Wilson–Delk): fully inflate, clamp tubing, bend opposite curve 60–90 s, hand on shaft not glans (urethral perforation). Straightens 86–100%.
- Residual >30° or buckling → relaxing incision (deflated; cautery <30 W with Titan); graft if defect >2 cm (never local dermis — infection).
- Residual <45°: home modelling from 4 weeks. Length loss commonest complaint. → *Ch 75*

7 · Landmark trials

Trial	Design	Result	Take-home
IMPRESS I and II	Phase III RCT, CCH vs placebo 2:1, curvature 30–90°	Curvature ↓34% (17°) vs 18% (9.3°); bother improved	Basis of FDA approval; modelling essential
Interferon α-2b multicentre RCT	Intralesional IFN vs saline	↓27% vs 9% curvature	First RCT of intralesional therapy

★ Putting it together: andrology decisions

A · Infertile man: work-up

ALGORITHM

1. History/exam (both partners): toxins, testosterone use, mumps, cryptorchidism; testis volume, vasa, varicocele (→ *Ch 69*).

2. Two semen analyses (abstinence 2–7 days). Low volume + acid pH → EDO or CBAVD; post-ejaculate urine (retrograde).

3. Hormones (low count/sexual dysfunction): FSH, testosterone (low → LH, prolactin).

4. Genetics (azoospermia/severe oligo): karyotype, Y microdeletions; CFTR in CBAVD (+ partner).

B · Azoospermia: obstructive or not

	Obstructive (OA)	Non-obstructive (NOA)
Testis	Normal volume	Small, soft
FSH	Normal (≤7.6 IU/L)	Raised (>7.6 IU/L)
Clues	Absent vasa, low volume, vasectomy/hernia repair	Klinefelter, Y microdeletion, cryptorchidism, chemotherapy
Treatment	VV/VE, TURED, or sperm retrieval for ICSI (→ *Ch 70*)	Micro-TESE + ICSI; fix clinical varicocele first; no retrieval if complete AZFa/b deletion

KEY POINT

† Reversal if younger partner, >1 child wanted; ICSI if older partner/female factor. Vas fluid copious (even clear, sperm-free) or sperm/parts → VV; toothpaste or none → VE (→ *Ch 70*).

C · Erectile dysfunction ladder

ALGORITHM

1. Assess: IIEF-5/SHIM, exam, HbA1c, lipids, morning testosterone. ED = cardiovascular sentinel; stratify cardiac risk (→ *Ch 72*).

2. Modify risk factors, drugs; treat hypogonadism (rescues PDE5I failures).

3. Shared decision: PDE5I (never with nitrates; re-counsel before failure), vacuum, intraurethral alprostadil, ICI (alprostadil/trimix).

4. Penile prosthesis (inflatable three-piece) if others fail; highest satisfaction. Infected: remove all ± salvage (→ *Ch 75*).

Young, focal post-traumatic arterial occlusion → revascularisation.

D · Priapism: ischaemic vs non-ischaemic

	Ischaemic (emergency)	Non-ischaemic (high-flow)
Clinical	Rigid, painful	Semi-rigid, painless; perineal trauma
Cavernous blood gas	pO_2 <30, pCO_2 >60 mm Hg, pH <7.25	Arterial: pO_2 >90, pCO_2 <40

	Ischaemic (emergency)	Non-ischaemic (high-flow)
Doppler	Little/none	High flow, fistula/pseudoaneurysm
Treatment	Aspirate ± irrigate → phenylephrine (100–500 µg every 3–5 min, up to 1 h) → distal shunt ± tunnelling → proximal shunt; >36 h or refractory → early prosthesis	Observe; selective embolisation if persistent

KEY POINT

Sickle cell: treat as above (no waiting for transfusion), no chilled irrigation, involve haematology. Stuttering: daily PDE5I, nightly pseudoephedrine, or post-pubertal hormonal suppression (→ *Ch 73*).

E · Peyronie disease

ALGORITHM

Active: analgesia, traction. CCH: stable dorsal/lateral 30–90°, intact erections; else verapamil, interferon α-2b.

Surgery: ≥12 months from onset, stable ≥6 months; document SPL, erections (→ *Ch 76*).

Good erections: simple <60–70°, no hourglass/hinge, adequate length → plication; complex, hourglass/hinge, short → incision/partial excision + graft (more ED). **Poor erections** despite PDE5I → inflatable prosthesis ± modelling.

F · Premature ejaculation (PE)

KEY POINT

Diagnosis: latency (lifelong ≈1 min or less) + inability to delay + distress, never latency alone. Treat ED first. Options: stop–start/squeeze, lidocaine–prilocaine spray, on-demand dapoxetine (only licensed oral outside US), daily SSRI (paroxetine best), tramadol (off-label, addiction) (→ *Ch 74*).